Caption Once, Ask Cheaply: What a Question-Blind Caption Index Buys a Video Reader

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Abstract

Video QA systems can caption a video once, without the question, and answer later questions from this caption index with a cheap text model. We ask why such an index can beat the frames it was made from, and what it saves. We give the captioner exactly the reader's frames, so the captions hold nothing the frames lack, and split their effect into source, visual budget, captioner and modality. In a pre-registered test on 297 TempCompass questions, Gemini's captions of Claude Haiku's frames raise Haiku's accuracy by 0.15 over its own captions. The gain recurs, in follow-ups predicted before they ran, at matched visual tokens, with Sonnet captioning, and with an open-weights reader. No reader gains measurably from its own captions over its frames, and two of three lose. The saving comes from the cheaper reader. Haiku reading a stronger model's index costs 1/23 to 1/26 of Sonnet reading the frames per question, repays captioning within four questions per video, and loses at most 9 points. Most wrong answers come from captions that leave the question undecided, largely on stress-test videos. Where index and frames disagree, the question's aspect and the models moderately predict which is right on new questions.

1 Introduction

Many video question answering systems first caption the video and then reason over the text (Zhang et al., 2024; Wang et al., 2025; Zhang et al., 2026). Captions written once, before any question is known, act as an index that every later question can reuse, so captioning is paid once per video and each question costs only a text read. Recent work builds such indexes for long videos (Cai et al., 2026; Liang et al., 2026; Chen et al., 2026), and in CapMem a model's own captions beat its frames only on videos longer than 20 minutes.

Two questions about such an index are still open. The first is why a reader would answer better from captions than from the frames they describe. The captioner may have seen more of the video, spent more visual tokens, or been a stronger model, or the reader may simply find text easier than pixels. The second is whether the index pays for itself. That depends on the whole cost of each question and on the one-time cost of captioning.

We hold the information fixed by giving the captioner exactly the frames the reader would see. A caption made from these frames cannot contain information about the video that the frames lack (Proposition 1). Any accuracy gain from the captions is therefore information that the reader can use more easily, not new information. Arms that change one thing at a time then split the effect of captions into four parts: the source the captioner sees (the video or the reader's frames), the visual tokens it spends, the captioner model, and the modality (a reader's self-captions against its frames). Claude Haiku and Sonnet read, Haiku, Sonnet and Gemini write captions, and an open-weights model, Qwen2.5-VL-7B, reads the main arms and captions for itself. The questions are 297 multiple-choice questions from TempCompass (Liu et al., 2024), on videos with a median length of 10 s. We set the main test on a separate development set and pre-registered it, and we logged a predicted value for each follow-up before running it.

We find that the captioner matters and the modality does not. With Haiku reading, Gemini's captions of the reader's frames beat Haiku's self-captions by 0.15 in the pre-registered test. The gain holds when both captioners spend matched visual tokens, and when Sonnet captions input identical to Haiku's. Haiku is no more accurate on its self-captions than on the frames they describe, Sonnet is less accurate on its own, and an open-weights reader replicates both results. Two of our pre-registered predictions failed: that Haiku would lose on its self-captions, and that captions would

lose direction and speed more than other aspects. The index pays through a cheaper reader. Haiku and Sonnet read Sonnet's captions within 0.02 of each other, although Sonnet is 0.19 more accurate on the frames. Per question, Haiku on a stronger captioner's index costs 1/23 to 1/26 of Sonnet on the frames, and it is 0 to 9 points less accurate.

Most of what captions lose is lost on videos that TempCompass built as stress tests. On unmanipulated videos, Sonnet on its self-captions is within 0.01 of its frames. When two copies of a clip play at different speeds, question-blind captions do not say which is faster, although the same model often sees the difference when it is shown the question. Captions written with the question in view recover most of this gap. On reversed clips the captions describe the forward event or state no direction, and writing them with the question recovers little. Where the index and the frames disagree, the question's aspect and the pair of models predict which one is right (AUC 0.67 on new questions about the same videos, for a rule frozen beforehand), though choosing by them gains little accuracy, and scores of the video's motion, its cuts or how well the captions cover the frames add nothing.

Our contributions are:

1. A decomposition of the effect of captions into source, visual budget, captioner and modality effects, with arms that change one thing each, in two model families (§3).

2. A pre-registered test and predicted replications, at identical input and with an open-weights reader, that show a positive captioner effect and a modality effect of at most +0.09 (§4.1–4.2).

3. A cost account per question that counts input, output and thinking tokens, with the break-even number of questions per video, also projected to longer videos (§5).

4. An account of what question-blind captions lose, from minimal pairs, question-aware controls and an audit of wrong answers (§4.3–4.5).

5. A decision rule for reading the index or the frames, frozen and tested on 738 new questions, and a null result for scores of the video and its captions as predictors (§4.6).

We study only short videos, multiple-choice questions and one benchmark.

2 Setup

2.1 Task and data

We use TempCompass's multiple-choice questions (Liu et al., 2024) on five aspects: action, speed, direction, order and attribute change. We set the hypotheses on a development set of 100 questions. The confirmatory set samples 60 questions per aspect, one per video, on videos the development set does not use, and 297 remain after three exclusions (Appendix C). TempCompass builds most of its videos to defeat shortcuts, so we report four video types: plain (137), reversed (74), stacked (41) and two-clip (45). A stacked video shows one clip twice, one copy above the other, at two speeds. The type nearly determines the aspect. The questions fall into 156 clusters of videos that share a filmed clip. We split each video into 5 s windows and take one frame per second at 224 px on the long edge (40 Claude image tokens each). The videos are short (median 9.7 s), and 58% have one or two windows. We keep the per-window design because it scales to long video, and one caption per video does no better (§4.7).

2.2 Arms

An arm gives the reader the question and options and, for each window, one kind of evidence: nothing (question only), the middle frame, all 5 frames, or one caption of those frames. Table 3 in Appendix A lists every arm with its visual tokens. The readers are Claude Haiku 5.5 and Sonnet 5.5 (pinned model IDs, effort medium), called through `claude -p` with no tools and a fresh session per call. As an open-weights check, Qwen2.5-VL-7B-Instruct (Bai et al., 2025) (float16, greedy decoding) reads the main arms (§4.7). Haiku, Sonnet, Gemini 3.8 Flash and Qwen write captions. We call Sonnet and Gemini *stronger* captioners than Haiku, Sonnet because it reads the same frames better (0.798 against Haiku's 0.613), and Gemini, which never reads, only because its captions are more useful (§4.1). A reader's *self-captions* are its own captions of the frames it would read. Every caption is question-blind and sees only its window, with two exceptions: the whole-video captions see all of a video's frames (§4.7), and the question-aware captions see the question (§4.5). All captioners get the same prompt (Appendix A).

2.3 Statistics and pre-registration

We set the main test, P1–P3, on the development set and pre-registered it: we wrote it down with its analysis code and committed both to the repository before any confirmatory call. It carries 98.33% CIs (Bonferroni over three). Every other comparison is exploratory. Each was written down with a predicted value and committed before its first call, its analysis was run once, and it carries a 95% CI. The two tests of the decision rule (§4.6) each have two primary hypotheses, at 97.5% (Bonferroni over two). Appendix B lists every term with its prediction. CIs are paired bootstrap over questions, and "cl." marks a CI from resampling the 156 clip clusters instead. The two rarely disagree (Appendix F). The repository's full history, which holds each prediction in a commit dated before its run, is archived by Software Heritage (Appendix I).

3 Framework

Proposition 1 (garbling). Let F be the reader's frames, Q the question and A its answer. A question-blind caption C of the frames is drawn by the captioner from F alone, so any answering rule that reads (Q, C) can be run on (Q, F) by first drawing C. Hence no rule on the caption is more accurate than the best rule on the frames, and $I(A; C \mid Q) \leq I(A; F \mid Q)$ by the data-processing inequality (Cover and Thomas, 2006). A bounded reader can still extract more *usable* information (Xu et al., 2020) from a good caption than from the pixels. The captioner and modality effects measure such differences.

The four effects. Each effect compares two arms that differ in one thing, with Haiku reading (Figure 1). *Source*: Gemini's captions of the video window against its captions of the reader's frames played as a 1 fps clip. Gemini samples the video at 1 fps too, at the same instants, so the source effect measures what the original file adds at those instants (mostly resolution), not motion between them. *Visual budget*: one captioner's captions of the same frames at two visual-token counts. *Captioner*: two captioners' captions of the same frames at equal or matched visual tokens. *Modality*: the reader on its self-captions against the reader on those frames (the two prompts also differ in the sentence that says what the evidence is; Appendix A). In the Claude chain, source and visual budget are zero by design. In the Gemini chain both branch from the frame-clip captions, so the terms add up along two paths (Figure 1).

Proposition 2 (break-even). An index τ that costs $C(\tau)$ per video and c_τ per question pays for itself, against frames at c_f per question, after

$$Q_{\mathrm{BE}} = \frac{C(\tau)}{c_f - c_\tau}$$

questions per video, the break-even quantity. With a reader downgrade, c_τ is the cheaper reader's cost. If each of W windows is captioned in its own call at cost k, and costs the reader s_f as frames and s_τ as text, then $Q_{\mathrm{BE}} \to k/(s_f - s_\tau)$ as W grows, so on long videos the break-even is set by the per-window prices and no longer by the fixed prompt.

Omission. Suppose the same model answers correctly from the frames with the question in view, but its question-blind captions do not decide the question. Then the captioner could see the fact and did not write it down, and we call the loss an omission.

Retention, (captions − question only) / (frames − question only), is the share of the frames' gain over the question alone that an index keeps.

4 Results

Unless stated otherwise, results are on the 297 confirmatory questions, and no question is missing from any arm. Differences are in accuracy, so +0.064 is 6.4 points.

Table 1 lists the main arms, and Table 10 the variants. Two rows anchor the rest. With the question and options alone, both readers score 0.41 (Haiku 0.411, Sonnet 0.414; difference +0.003 [−0.061, +0.064]), well above chance (≈ 0.30), so the two readers share a strong prior from the wording. Given five frames per window, Sonnet scores 0.185 [0.131, 0.239] more than Haiku (0.798 vs. 0.613). Because both readers start from the same floor, this gap comes from reading the input rather than from prior knowledge of the questions. Every index Haiku reads lands between Haiku's and Sonnet's accuracy on frames (0.643–0.791), and every index Sonnet reads lands below its frames (0.710–0.791).

4.1 Who writes the caption matters

The pre-registered test of the captioner is P3. With Haiku reading, Gemini's captions of exactly the reader's frames beat Haiku's self-captions by +0.148 [+0.088, +0.209], 2.5 times the +0.06 we predicted. The gain is positive in every aspect

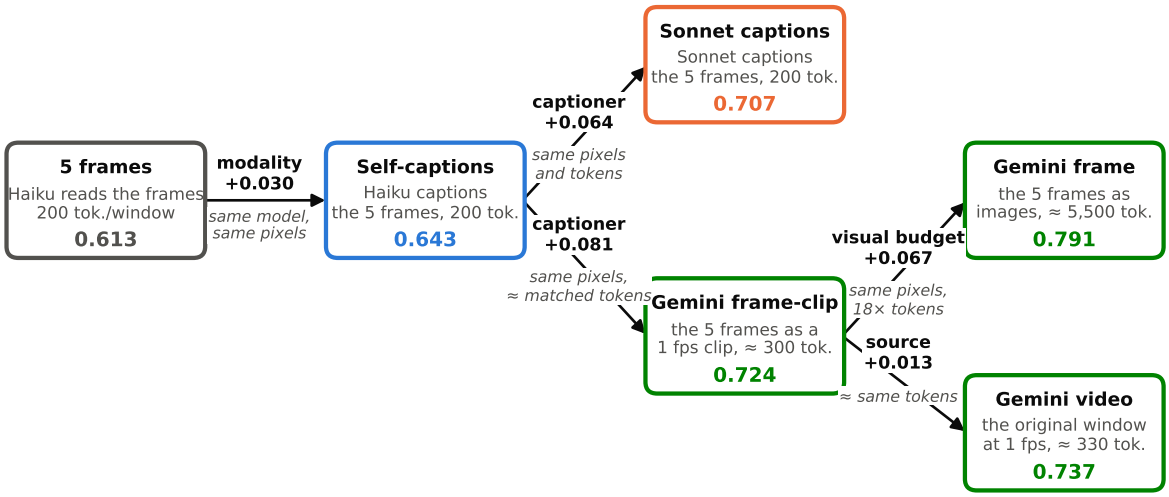


Figure 1: The design. Haiku reads every box (accuracy on the 297 confirmatory questions); the colour marks the captioner. Each arrow changes one thing, and its label is the effect, the accuracy difference between its two boxes, with what the arrow holds fixed below it. The effects add up along each path, e.g. Gemini frame − 5 frames = modality + captioner + visual budget. CIs are in §4.1–4.2 and Table 7 (Appendix E).

Reader, input	Acc.	In tok.	$/q	Index
H, question only	0.411	1,025	1.09	—
H, 5 frames	0.613	1,520	2.04	—
H, self-captions	0.643	1,283	1.89	0.58
H, Sonnet capt.	0.707	1,280	1.81	9.10
H, Gem. frame-clip	0.724	1,232	1.67	7.60
H, Gem. frame	0.791	1,229	1.67	14.61
H, Gem. video	0.737	1,236	1.68	7.26
S, question only	0.414	1,027	21.82	—
S, 5 frames	0.798	1,522	41.12	—
S, Sonnet capt.	0.710	1,282	30.89	9.10
S, Gem. video	0.791	1,238	28.68	7.26

Table 1: Accuracy and list-price cost per question (297 questions; H = Haiku, S = Sonnet reading). Input tokens are the median per question and include the fixed reader prompt (≈ 1,025 tokens). $/q is list-price dollars per question $\times 10^4$, including output and thinking tokens. Index is the one-time captioning cost in dollars per video $\times 10^3$, averaged over videos (§5). The variant arms and Qwen's arms are in Table 10.

(+0.10 to +0.20) and in every video type (Appendix D).

P3, however, compares two captioners that spent different numbers of visual tokens on the same pixels. Gemini's image path encodes each 224 px frame at ≈ 1,100 tokens, ≈ 5,500 per window, against Haiku's 200. Captioning the same frames as a 1 fps clip through Gemini's video path (≈ 280–330 tokens per window, as in the video arm) splits P3 into two parts of similar size. The captioner effect at about matched visual tokens is +0.081 [+0.030, +0.131], cl. [+0.035, +0.125], and the visual budget effect is +0.067 [+0.030, +0.108]. Claude's captioner shows no such budget gain in the range we tested. Sonnet's captions of 448 px frames, at 3.6 times the visual tokens, are no more useful to Haiku (+0.003 [−0.037, +0.047]; Appendix F).

The source effect is near zero (+0.013 [−0.030, +0.061]). Gemini samples the video window at the same 1 fps instants and reduces each frame to ≈ 66 tokens, so little of the original file's resolution reaches the captioner, and nothing between the instants does. All four predictions for this split were within 0.02 of the results.

Within one model family, we can match the captioner's input exactly. Sonnet captions the same five JPEGs per window, with the same prompt and effort as Haiku's self-captions, and its caption calls took exactly 2 more input tokens than Haiku's in each of the 876 windows. Only the captioning model differs. Haiku reads Sonnet's captions more accurately than its self-captions, by **+0.064 [+0.017, +0.111]**, cl. [+0.011, +0.119], and the predicted +0.09 lies inside the CI. The effect also holds on minimal pairs, where a guess cannot score (+0.104 [+0.017, +0.190] in pair accuracy; §4.3), on 738 further questions about the same videos (+0.072; §4.6), and with Qwen reading its self-captions and Sonnet's (+0.125 [+0.074, +0.175]; §4.7). At about equal visual tokens, Sonnet's and Gemini's captions do not differ measurably in how useful they are to Haiku (Sonnet − Gemini frame-clip captions, −0.017 [−0.071, +0.037]).

4.2 Text itself is not easier for the reader

The modality effect asks whether a reader does better on its own words than on the pixels they came from. For Haiku this is the pre-registered P2, self-captions − 5 frames, which comes out at +0.030 [−0.034, +0.094]. Both our prediction (−0.08) and any gain above +0.09 lie outside the CI. On 738 further questions about the same videos the effect is −0.064, cl. [−0.114, −0.014] (predicted +0.03), because Haiku's frames score higher there, mostly on direction (§4.6). The sign for Haiku thus depends on the questions, and neither set shows a gain from its own words. Sonnet and Qwen lose accuracy on their self-captions. For Sonnet the effect is −0.088 [−0.135, −0.044], cl. [−0.140, −0.036], or −0.067 [−0.114, −0.024] when it captions the 448 px frames, and for Qwen it is −0.121 [−0.172, −0.071], cl. [−0.170, −0.074]. The modality effect is thus near 0 only for Haiku, the reader with the lowest point estimate on frames (Qwen − Haiku +0.044 [−0.010, +0.101]; Table 10).

The readers tie on Sonnet's captions. From Haiku to Sonnet, accuracy on frames rises from 0.613 to 0.798, while accuracy on Sonnet's captions stays flat (0.707 → 0.710; Sonnet − Haiku +0.003 [−0.040, +0.047]; Figure 2 in Appendix E). On these captions, a reader that costs 17× less per question is not measurably less accurate. The tie holds on all four indexes Sonnet wrote (Sonnet − Haiku +0.000 to +0.020, every CI includes 0; Appendix F). Haiku on frames is below what the captions hold and gains from them, and Sonnet is above it and loses. A fixed compressed text narrows the gap between readers in retrieval-augmented QA too (Panthi and Abdelfattah, 2026). A Sonnet-written index is also in Sonnet's own words, so we predicted the readers' gap on Haiku's self-captions too and had Sonnet read them. Sonnet is not measurably better than Haiku on these either (+0.034 [−0.010, +0.081]), and the two gaps do not differ measurably (−0.030 [−0.088, +0.027]). The tie therefore does not come from Sonnet reading its own words, although we cannot exclude an advantage of up to 0.08 on Haiku's captions.

The tie does not hold for every index. Sonnet reads Gemini's video and frame-clip captions more accurately than Haiku does (+0.054 and +0.064), and on Gemini's frame captions it is level with its frames (0.811 against 0.798). Qwen trails Haiku on every shared index (−0.04 to −0.08), mostly on two-clip videos (Appendix E). Haiku's advantage on captions does not come from starving the frames, since captions still beat 448 px frames at 1.76× the input tokens (Appendix F).

Arm	All	Rev. (107)	Stack. (41)	2-clip (45)	Same ans.
H, 5 frames	0.233	0.28	0.17	0.18	0.54
H, self-captions	0.326	0.33	0.10	0.53	0.42
H, Sonnet capt.	0.430	0.38	0.17	0.78	0.43
H, Gem. video	0.482	0.47	0.20	0.78	0.37
S, 5 frames	0.606	0.56	0.51	0.80	0.28
S, Sonnet capt.	0.430	0.40	0.10	0.80	0.39

Table 2: Minimal pairs: pair accuracy by group, and the share of pairs with the same answer on both videos. Chance pair accuracy is 0.093.

4.3 Where captions lose: video types and minimal pairs

Built videos (reversed, stacked and two-clip) make up 54% of the confirmatory set. Every stacked question is about speed and every two-clip question about order, so we report results by type (Figure 3 in Appendix E). On plain videos, the five arms lie within 0.09 of each other (0.80–0.89). They separate on the built videos.

Accuracy on one video cannot separate a representation that lost the fact from a reader that guessed, but minimal pairs can. We ask each of 193 confirmatory questions again, with the same wording and option order, on a sibling video built from the same filmed clip whose correct option differs. A pair is the plain and the reversed version of a clip (107 pairs), two stacked versions at different speeds (41), or two arrangements of the same two clips (45). An arm that does not receive the fact cannot tell the two videos apart, and if it gives the same answer on both, it gets at most one right. Table 2 reports pair accuracy, the share of pairs answered correctly on both sides (chance 0.093), and the same-answer rate.

Captions help a weak reader on two-clip videos. From frames, Haiku must keep about ten small frames from two clips apart, and on two-clip pairs it is near chance (0.18). Its self-captions raise its pair accuracy to 0.53, a modality effect of +0.36 [+0.16, +0.56], and Sonnet's captions raise its accuracy over its frames by +0.33, cl. [+0.20, +0.48]. We predicted both gains.

On stacked videos, question-blind captions drop the comparison. Sonnet on frames answers 0.51 of stacked pairs correctly on both sides. Every cap-

tion arm scores 0.10–0.20, near pair chance, including Sonnet on its self-captions (0.10; −0.41 [−0.62, −0.23] against its frames). For a man jumping rope in two copies (answer: the one above is faster), Gemini writes "An athletic man … jumps rope in a split-screen video". Sonnet often sees the difference when it looks with the question in view, and loses it when it writes without the question. Much of the loss is thus an omission (§3), and captions written with the question in view recover most of it (§4.5).

On reversed videos, captions narrate forward or do not commit. The question alone points to the forward event, so from the question alone both readers score below chance on reversed videos (Haiku 0.16, Sonnet 0.18). Sonnet on frames recovers part of the reversal (0.64), but its question-blind captions keep little of it (−0.18 against its frames), and its question-aware captions keep barely more (−0.15; §4.5). For a pile of salt played backwards (answer: shrinking), all five of Gemini's video captions describe salt pouring onto the pile. Split by type, the loss from writing blind sits on reversed and stacked videos, and on plain and two-clip videos Sonnet's captions keep 0.95–1.00 of its frames' gain (Appendix E).

4.4 Why caption arms fail: an audit

We audited every caption set and question that a reader got wrong in one of seven reader arms (442 items). A judge (Sonnet) saw the captions, question, options and correct answer. It labelled the captions as stating the correct option (a reader error), a wrong option (a misstatement), or neither (an omission), and quoted the caption as evidence (Appendix A). In every arm, most wrong answers (0.61–0.73) come from captions that do not decide the question, and reader errors are only 0.05–0.17 (Table 9 in Appendix E). No reader can read a fact that is missing from the index, which fits the readers' tie on Sonnet's captions (§4.2). The wrong answers are concentrated on built videos, which hold 54% of the questions but 0.74–0.84 of each arm's wrong answers. One author labelled 40 items, blind to the judge's label, and agrees with the judge at $\kappa = 0.81$ [0.64, 0.96].

4.5 From question-blind to question-aware

Three further arms move the description toward the question, with new captions (Appendix E). When Haiku sees the frames and the question and *describes, then answers* (one sentence per window, then the answer, in one call), it scores 0.684, +0.071 over the frames arm, cl. [+0.029, +0.114]. On stacked videos it beats its question-blind self-captions by **+0.22 [+0.09, +0.37]**. That arm reads the frames at every question and is not an index. *Question-aware captions*, written from the same 5 frames with one added bullet that gives the question and options, and read without the frames, test whether text alone can carry what blind captions omit. Overall they gain little over the blind captions (+0.017 for Haiku and for Sonnet, each reading its own; predicted +0.04 and +0.05). On stacked videos Haiku gains +0.22 [+0.07, +0.39], and Sonnet's gap to its frames shrinks from −0.27 to −0.10 [−0.28, +0.05], while on reversed videos the gap barely moves (−0.18 to −0.15). Text written with the question thus carries most of the stacked comparison, and the reversal is lost even when the captioner knows the question. A bullet that names only the five kinds of question gains +0.040 [+0.003, +0.081] over Sonnet's whole-video captions with Sonnet reading and +0.027 [−0.013, +0.067] with Haiku, and on stacked videos Sonnet on frames stays 0.27 ahead.

4.6 When to read the index: a decision rule

A system that keeps both the index and the frames could choose between them per question. A reader's answers from the two disagree on 20% of questions, and either can be right. In three logged steps (Appendix G), we asked whether anything known before either answer is read says which will be right.

Scores of the video and its captions do not predict. *Motion* (the mean SigLIP 2 distance between consecutive frames; Tschannen et al., 2025) and *coverage* (the SigLIP 2 similarity of each caption to its window's frames) fail as predictors of a *loss*, a reader right on its frames and wrong on its self-captions, within each video type. The stratified AUC is 0.42 [0.28, 0.57] for motion (predicted 0.60) and 0.53 [0.38, 0.67] for coverage (predicted 0.58; 97.5% CIs over clip clusters).

A rule from the aspect and the models. In a development run, we fit logistic models to the 898 disagreements of 15 reader–index pairs. The question's aspect and the pair (reader, captioner family) give an out-of-fold AUC of 0.756 [0.705, 0.803]. Adding motion, cuts, the number of windows, coverage, or how often the captions use the aspect's

words ("faster", "leftward", "then") raises it by at most 0.002 (Table 13). We froze this rule and tested it on the 738 TempCompass questions about the confirmatory videos that the confirmatory set does not ask (2,952 Haiku calls). In words, the rule reads Gemini's captions, and with Claude's captions it reads the frames on speed questions. On the 492 disagreements its AUC is **0.666** [0.587, 0.743] (97.5%, cl.; predicted 0.68). The second primary test, a shift toward the frames on speed questions, narrowly fails (-0.148 [-0.320, +0.007]; predicted -0.25). The rule beats always reading the index by +0.013, cl. [+0.002, +0.026], but not always reading the frames (+0.014 [-0.013, +0.041]). The test reuses the confirmatory videos, and on the 67 disagreements about an aspect the confirmatory set had not asked of the video, the rule does not hold (AUC 0.40, descriptive). On long video, Anonymous (2026b) instead look at the frames when two text answers disagree, a signal at answer time that our rule does not use.

4.7 Robustness checks and nulls

Qwen2.5-VL-7B-Instruct reads eight arms and captions every window from exactly the 5 frames. Its self-captions are the weakest in the study, even for itself, and Gemini's video captions add only +0.040 [-0.010, +0.091] over its frames (predicted +0.12; Appendix F). One Sonnet caption per video instead of one per window does no better (+0.013 with Haiku reading, +0.010 with Sonnet), at half the index cost. Re-captioning identical input changes 10–17% of answers, so comparisons between caption arms, and by type, are noisy. When Haiku re-read its own self-captions 13 days later, 10.8% of its answers changed, with no measurable change in accuracy (-0.013 [-0.047, +0.020]). The preregistered aspect contrast (P1) did not pass. It is -0.101 [-0.231, +0.028] for Sonnet, half the predicted -0.20.

5 Cost and break-even

Accounting. We price every call at list price: Claude Haiku 5.5 at \$0.10 / \$0.50 and Sonnet 5.5 at \$2 / \$10 per million input / output tokens, with thinking billed as output (this equals the CLI's reported cost on every call), and Gemini captioning at \$0.50 / \$3.00. Q_{BE} follows Proposition 2, with c a reader's mean cost over the 297 questions and $C(\tau)$ the cost of captioning all of a video's windows once. Its CIs resample questions, and thereby videos, since each video has one question.

For one reader, the index saves little. Against 5 frames, Gemini video captions save 18% per question for Haiku and 30% for Sonnet. The percentages are small because the fixed reader prompt ($\approx$ 1,025 tokens) dominates the input (Appendix F). Sonnet on Gemini video captions repays after $Q_{\mathrm{BE}} = 5.8$ [5.2, 6.5], with no measurable accuracy difference (-0.007 [-0.054, +0.040]). Haiku repays only after 202 [156, 279], though it is 0.125 more accurate than on its frames.

The reader downgrade pays. Per question, Haiku on any index written by a stronger captioner costs 1/23 to 1/26 of Sonnet on frames (Table 1). The index repays its captioning within 1.2–3.7 questions per video, 1.2–1.3 for the whole-video indexes and 1.8–3.7 for the per-window ones (Table 11 in Appendix F). At 10 questions per video (Figure 4 in Appendix F), every index Haiku reads costs less than half of Sonnet on frames, captioning included. How much accuracy the downgrade costs depends on the index. Gemini frame captions bring Haiku within 0.01 of Sonnet on frames (-0.007 [-0.057, +0.044]), but they are the most expensive index (\$0.0146 per video). In the Claude chain, where Sonnet captions once and Haiku answers, the downgrade costs -0.091 [-0.145, -0.040] at $Q_{\mathrm{BE}} = 2.3$, and aspect-aware captions reduce this to -0.051 [-0.104, +0.000] at $Q_{\mathrm{BE}} = 1.3$. The CLI adds $\approx$ 850 input tokens to every Claude call, which an API deployment would drop or cache. Without them, the reader downgrade costs 1/25 to 1/34 of Sonnet on frames per question and repays within 1.4–6.3 questions per video (Appendix F).

Longer videos. Projected by Proposition 2 to longer videos whose every window is read (Appendix F), captions save more per question as videos grow, but Q_{BE} does not fall, because captioning also grows with every window. For the reader downgrade, Q_{BE} rises from 1.8–3.7 to 4.5–8.9 at one minute. On 39 three-minute EgoSchema videos (36 windows) from a development run, with their larger frames counted, the measured costs per question are 1.04–1.11 times the projected ones, and the measured Q_{BE} 0.93–1.00 times.

6 Related work

Caption, then reason. Composing models through language goes back to Socratic Models (Zeng et al., 2022), and many video QA systems caption first

and reason over the text (Zhang et al., 2024; Wang et al., 2025; Zhang et al., 2026; Wang et al., 2024a), or let an LLM agent choose which frames to caption (Wang et al., 2024b); LLoVi already finds that the captioner matters. Caption indexes and memories reuse the captions across questions (Cai et al., 2026; Liang et al., 2026; Chen et al., 2026; Goulas et al., 2025). Caption utility is also judged by QA rather than by similarity (Yang et al., 2026; Wang et al., 2026b; han Li et al., 2026), and CapRL (Xing et al., 2025) trains a captioner on it. Prism (Qiao et al., 2024) and MetaCaptioner (Lei et al., 2025) vary the captioner in front of a fixed reader. CapMem (Liang et al., 2026) and REEL (Liang and He, 2026) are the closest work. On long egocentric video, CapMem's Qwen readers read the captions of six Qwen captioners of the same frames, and their own, but it does not separate source from visual budget, cross model families or price a cheaper reader. In CapMem the stronger reader gains on fixed captions. On Sonnet's captions we find no such gain, and on Gemini's a small one. REEL gives a model's own question-blind text record and its cached visual states exactly the frames its frame path sees, on long video with six open models, and the text record changes 19.7–28.7% of answers at nearly equal mean accuracy. One model writes and reads, so REEL has no captioner swap and no cheaper reader. Our control extends theirs to a captioner other than the reader.

Text or pixels. IsoBench (Fu et al., 2024) finds that models score 15–29 points higher on exact text renderings of images than on the images, and TokenSwap (Hua et al., 2026) studies the same contrast. When a model reads its own description, the sign varies. Own captions give a weak 7B model +22 to +29 points on synthetic image tasks (Weng et al., 2025), while appending the model's own caption costs about 10 points on HallusionBench (Li and Zhang, 2026), and a one-call "state first" prompt does not help on charts (Wang et al., 2026a). On Visual7W, each of seven open models is less accurate on its own question-blind caption alone than on the caption with the image, by 11.3 points on average (Anonymous, 2026c). Our modality effect, at most +0.09 for Haiku and negative for Sonnet and Qwen, has the sign CapMem finds on videos under 20 minutes.

Readers on a fixed text. In retrieval-augmented QA, when every reader gets the same compressed text per question, the gap between the weakest and the strongest reader on HotpotQA shrinks from 31.8 to 7.8 points with one compressor, and weaker readers gain more (Panthi and Abdelfattah, 2026). Our readers' tie on Sonnet's captions is the same effect for a video caption index.

Temporal video, routing and cost. Our minimal pairs follow TempCompass's conflicting videos (Liu et al., 2024), MotionBlind (Bhatia et al., 2026) and TimeBlind (Li et al., 2026), but we apply them to representations instead of models. Video models often fail direction, speed and reversal from pixels too (Lee et al., 2026; Xue et al., 2025; Rahman, 2026; Li et al., 2024). Cascades and routers send each query to a cheaper model when they can (Chen et al., 2023; Ong et al., 2025; Dwivedi et al., 2026), and an index opens that cheap route for video. We found no paper that solves for the break-even with a reader downgrade. Appendix J covers further work on routing, on what an index keeps, and on cost.

7 Discussion and conclusion

We gave each captioner exactly the reader's frames. A stronger captioner's question-blind captions then make Claude Haiku and an open 7B reader more accurate, while no reader gains measurably from its self-captions (at most +0.09, for Haiku), so an index should be written by a stronger model than its reader. The saving comes from the reader downgrade. With the same reader, an index saves only 18–30% per question, while a cheaper reader, which on Sonnet's captions is within 0.02 of Sonnet, cuts the cost by a factor of 23–26 and repays the captioning within a few questions per video. A blind index loses what the captioner does not write without the question, namely comparisons between two copies of a clip and reversals, so a system should keep the frames for such questions. The question's aspect and the models moderately predict whether the index or the frames will be right on a question (AUC 0.67 on new questions about the same videos), and scores of the video and its captions add nothing. Without self-caption, frames and question-only arms, a "caption effect" would mix the captioner, the modality and the prior, since the same captions help Haiku and cost Sonnet (retention 1.47 and 0.77).

Limitations

Scope. We test one benchmark, short clips (median 9.7 s) and multiple-choice questions. Of the confirmatory videos, 54% are manipulated (reversed,

stacked, two-clip), so overall accuracies mix natural video with stress tests. We report plain videos separately (Table 8). The costs for longer videos (§5) are projections from short clips, for a reader that reads every window. We checked them once, on 3-minute videos, where they held within 11% once frame size was counted. All captions except the question-aware ones (§4.5) follow one generic, question-blind prompt that does not mention the benchmark's manipulations. Captions written for retrieval or question answering may keep more.

Contamination. TempCompass (March 2024) predates every reader's training cutoff (June 2026 for the Claude models), so the readers may have seen its questions. The question alone scores 0.40–0.41, 0.11 above chance, but the excess comes from 100 of the 297 questions. For 197 of the 297 confirmatory questions, TempCompass itself asks the same question, with the same options, of another video with another answer, so a remembered answer cannot fit both. On these 197 the question alone is at chance (Haiku +0.02, Sonnet −0.01), and the headline effects keep their signs: captioner +0.08 [+0.02, +0.14], the readers' tie on Sonnet's captions (Haiku − Sonnet) +0.02 [−0.04, +0.07], Haiku on Sonnet's captions vs. Sonnet on frames −0.11 [−0.18, −0.04]. The captioner effect and the downgrade's cost are larger there than on the other 100. On minimal-pair accuracy, which a text-only answer cannot earn, they hold too (Table 2; the tie is 0.00 [−0.07, +0.06]). We did not extend the study to EgoSchema (Mangalam et al., 2023), because in a development run Sonnet scored 0.72 from the question alone (chance 0.20; 39 questions). A confirmatory EgoSchema run, logged in advance, was stopped for this reason 53 minutes after it started, before any of its results were computed.

Readers. The closed readers run through a CLI, so its system prompt is part of every input and temperature is not controlled. The CLI's built-in prompt changed between the first runs (October 18–20) and the last (October 31). A re-read of Haiku's self-captions across that gap changed accuracy by −0.013 [−0.047, +0.020] (§4.7). Costs are at list price for these calls, and §5 also reports them without the CLI's prompt. The model decides per call whether to think (effort medium), so we observe thinking but do not control it. Our mitigations are pinned model IDs, isolation checked on every call, released outputs, and an open-weights reader with greedy decoding that replicates the captioner and modality results (§4.7). That reader is one 7B model, with its own default system prompt. In the main chain Claude reads Claude's captions, and the Gemini chain and the Qwen reader check this. Haiku's accuracy on frames dropped from the development to the confirmatory set (0.70 to 0.61), and about half of the drop is the different mix of video types.

Inference. Questions share filmed clips, so we also report CIs over clip clusters. Most claims beyond the pre-registered test are exploratory. We predicted each follow-up in writing before it ran, but not the readers' tie on Sonnet's captions, the thinking pattern, the analysis by video type of the first two runs, or the checks marked as computed after their run. The audit's judge (Sonnet) also wrote one of the caption sets it judges, and 40 blind hand labels check it ($\kappa = 0.81$). We tested the decision rule (§4.6) on new questions about the confirmatory videos, not on new videos. Of the 738, 618 ask the same aspect of a video as a confirmatory question, mostly reworded, so the test is closer to a replication than to an independent sample. On the same videos and with the same number of options, the new direction questions moved Haiku's accuracy on frames by 0.20 (Appendix G). Accuracies for one reader thus depend on how the questions are worded as well as on the videos.

Question-blind captions drop what the question needs and the captioner did not write (stacked comparisons, reversals). Sonnet partly sees the stacked comparison on frames (0.66), and captions written with the question in view recover most of it (§4.5), so much of that loss comes from writing blind. They do not recover the reversals. Why a description of the frames loses a reversal that the same model partly sees in them remains open. The question-aware captions are written per question and are not an index. Whether a fixed index can keep these facts without knowing the question is untested beyond the aspect bullet.

Future work. Videos released after the readers' training cutoff, with answers made by editing them, would rule out contamination. A signal at answer time, such as two text answers that disagree (Anonymous, 2026b), may choose between index and frames where the aspect and the models do not. A task that asks which candidate fills a masked span would compare the modalities on reconstruction. Over 1 s gaps, copying the adjacent frame is level with a text reader (Appendix F), but the missing twenty minutes of a wedding, with its vows and

rings, may be easier to guess from script knowledge than from pixels. Long videos would show where captioning everything stops paying.

Ethics Statement

The videos and questions are TempCompass's (CC BY-NC 4.0). We release the captions under the same licence and do not redistribute videos or frames. No human subjects took part, and one author labelled 40 audit items. **Compute.** The TempCompass study made 17,962 Claude calls (12,798 reads, 4,721 captions and 443 audit labels; $28.22 at list price, run on a subscription), the EgoSchema development run 3,203 more ($7.94), and the stopped EgoSchema run 2,107 ($5.31). We spent $9.67 on Gemini captioning and ran Qwen on Kaggle's free GPUs (2×T4, 74 minutes for its captions and reads). Earlier phases of the project (Appendix F, forced choice) are not counted. **AI assistance.** Claude (through Claude Code) wrote analysis code, ran the experiments, and drafted and edited text under the author's direction. The author checked the reported numbers against the logged outputs and is responsible for the content.

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A Arms and prompts

All prompts are given exactly as sent. Text in braces is filled in per call.

Caption prompt. Every captioner gets the same prompt. For frames (Haiku, Sonnet, and Gemini's frame and frame-clip captions), the first sentence is:

```
The attached images are {n} frames from a {d}-second video
      clip, taken one second apart and shown in time order.
      Describe what happens in the clip, in one or two
      sentences of at most 45 words in total.
```

For Gemini's video captions, it is "Describe what happens in the attached 5-second video clip, in one or two sentences of at most 45 words in total." The rest is the same for all:

Arm: what the reader sees	Vis. tok./window
Question only: question and options	0
1 frame: the middle frame of each window	40
5 frames: all 5 frames of each window	200
5 frames, 448 px	720
Describe, then answer: 5 frames; one sentence per window, then the answer	200
Self-captions: the reader's own isolated caption of each window's 5 frames	— / 200
Sonnet captions: Sonnet's isolated caption of the same 5 frames	— / 200
Sonnet captions, 448 px: from the 448 px frames	— / 720
Sonnet whole-video captions: one caption of all of a video's frames	— / 200
Sonnet aspect-aware captions: the same, with one bullet naming the kinds of question	— / 200
Question-aware captions: Haiku's or Sonnet's, with one bullet giving the question	— / 200
Gemini frame-clip captions: the same 5 frames, played as a 1 fps clip	— / ≈ 280–330
Gemini frame captions: the same 5 frames, as images	— / ≈ 5,500
Gemini video captions: the video window, sampled by Gemini at 1 fps	— / ≈ 330

Table 3: Arms (§2.2). Visual tokens per 5 s window, for the reader / for the captioner.

```
* Start with what happens: what the people, animals or
      machines do, or, if nothing in the clip acts, how the
      scene or the camera view changes.
* Then add the details that set this moment apart: the key
      objects and any on-screen text, but do not transcribe
      subtitles, clocks, timestamps, dates or numbering (
      such as day, part or step numbers), and leave out
      watermarks and channel logos.
* Describe the setting in a few words at most.
* Mention camera movement only when the camera clearly moves.

* Describe what is seen, not what is heard.
* Include enough details so someone who hasn't seen the
      video could understand what's happening.
* If a description cannot be generated, write "ERROR".
```

The whole-video captions (§4.7) get all of a video's frames in one call, and the length clause scales with the number of windows (for two windows: "in at most 4 sentences of at most 90 words in total"). The aspect-aware captions (§4.5) add one bullet before the last:

```
* The description will be used later to answer questions
      about the actions, the speed of the movements, the
      direction of the movements, the order of the events,
      and changes in the attributes of things (such as size,
       colour, shape or state). State these whenever the
      frames show them.
```

The question-aware captions (§4.5) are per window, and add instead, before the last bullet:

```
* The description will be used, together with descriptions
      of the other parts of the same video, to answer this
      question about the video: {stem} The options are: {A.
      option; B. option; ...}. Describe what these frames
      show that bears on the question.
```

Reader prompt.

```
Answer a multiple-choice question about a video clip that is
      {duration} seconds long. There is no sound.

{modality}

{evidence}

Question: {stem}
{options}

Answer with JSON only, in this format: {"answer": "<letter
      >"}, with one of the letters {letters}.
```

The evidence is the frames or the captions, each window labelled with its time span ("0-5 s:"). The modality sentence is, for frames:

```
The video was split into windows of 5 seconds (the last one
      may be shorter). Each window is shown as its frames
      taken one second apart, in time order.
```

for captions:

```
The video was split into windows of 5 seconds (the last one
      may be shorter). Each window is given as a short
      description of what can be seen in it, in time order.
      Each description was written by an AI model that
      watched only that one window, without sound, and knew
      nothing about the rest of the video.
```

for whole-video captions:

```
The video is given as a single description of what can be
      seen in it. The description was written by an AI model
       that watched the whole video, without sound.
```

for one frame per window, "… Each window is shown as one frame from its middle, in time order."; and for the question only, "You cannot see the video. Give your best guess from the question and options alone."

Describe, then answer. The reader prompt with frames, with the last paragraph replaced by:

```
First describe each window in one sentence, in time order:
      what can be seen in it and what happens. Then answer.
      Answer with JSON only, in this format: {"windows": ["<
      window 1>", "<window 2>", ...], "answer": "<letter>"},
       with one description per window and one of the
      letters {letters}.
```

Audit judge (§4.4).

```
You are checking whether written descriptions of a video
      contain the answer to a question about it.

The video is {duration} seconds long, without sound. It was
      split into windows of 5 seconds (the last one may be
      shorter). An AI model wrote one description per window,
       seeing only that one window and not the question:

{captions}

Question: {stem}
{options}

The correct answer is {answer}. {answer_text}

Judge the descriptions only, not what you think the video
      shows. Choose one label:
- "correct": the descriptions state the correct answer, or
      imply it clearly enough that a careful reader would
      choose it.
- "wrong": the descriptions state, or clearly imply, a
      different option.
- "silent": the descriptions do not decide between the
      options: the fact needed is missing or too vague.

Answer with JSON only, in this format: {"label": "<correct|
      wrong|silent>", "evidence": "<a short quote from the
      descriptions, or an empty string>"}
```

B Development phase and predictions

Development phase. The hypotheses came from 100 development questions (20 per aspect, one per video, on videos the confirmatory set does not use), read by Haiku (question only, Gemini video captions, 1 and 5 frames per window) and Sonnet (Gemini video captions and 5 frames), and on 50 of them by Haiku with its self-captions. On them, Haiku on Gemini captions tied Haiku on 5 frames (+0.01 [−0.07, +0.09]), and Sonnet lost on the captions (−0.07 [−0.15, 0.00]), mostly on direction and speed. Haiku's self-captions were below its frames (−0.10 [−0.22, +0.02]) and below Gemini's captions (−0.08 [−0.18, 0.00]). These results gave P1 (captions lose direction and speed: the grouping direction, speed against the other three aspects was fixed before any confirmatory call), P2 (no modality advantage, predicted −0.08) and P3 (the captioner adds usable information, +0.06). The predictions set on the development set were too pessimistic about captions, and S1, S3, S5 and S6 all came out higher than predicted (Table 4). Part of the shift is the mix of video types. Built videos are 40% of the development set and 54% of the confirmatory set, and weighting the confirmatory accuracies by the development mix accounts for about half of Haiku's drop on frames (0.70 → 0.61).

Predictions against results. Every exploratory term was written down with a predicted value and committed before its first call. Table 4 lists every predicted term. Of the 73 accuracy differences in it, the prediction lies inside the result's CI for 58, and for 56 of the 64 after C1. Of the 65 non-zero predictions, 62 have the sign of the result. The three exceptions are P2 and S2, both set from the 50 development questions, and A8's modality term. The median absolute difference between prediction and estimate is 0.023. All 17 predictions of C13, C14 and A10 lie inside their CIs, and A9's cost checks fall within their stated bands. The misses after C1 are C4e (Sonnet's pair-accuracy loss on its self-captions, larger than predicted), three open-reader terms (C12a, d, f), whose predictions treated Qwen as a weaker Haiku (on frames it scores 0.044 above Haiku, CI including 0, and its captions are less useful; §4.7), and four A8 terms, which all missed in

the same direction because Haiku's frames were stronger on A8's questions than on the confirmatory ones (§4.6). Of the six AUC and speed-shift terms of A6 and A8, five have the prediction inside the CI (A6's gain over video type at its upper end). Of the four primary tests among them, only A8's AUC passes.

C Amendments, deviations and term mapping

Terms. The pre-registration called the modality effect the *format* effect, and the source effect *coverage*; they were renamed before any later entry, and the pre-registered entries keep their words. C1's coverage term (S2: Gemini video − frame captions) mixes the source with a 17× difference in visual tokens, so the paper's source effect is C2c, at about matched tokens.

Amendments, each made before the calls it affects.

- **Sampling (C1).** The first sampling rule (aspects filled rarest first) reached only 59 action and 59 direction questions and wrote no file; a preference for videos with a single aspect was added, and the sample was drawn once.
- **Gemini image cost (C1).** Gemini encodes each 224 px image at ≈ 1,085 tokens at its default media resolution, not the assumed ≈ 66, so a frame-caption call cost about 3× the planned amount; the spend cap was raised and the resolution kept, so the captioner settings stayed as tested.
- **Exclusions (C1).** Gemini returned no caption for 4 of 1,544 windows (0.26%, below the 2% stop rule): both windows of one video, blocked as prohibited content, and one window in each of two two-clip videos, where the window spans the cut and the frame captioner answered "ERROR". The 3 affected questions (all order) were dropped from every arm, leaving 297. Two of them are two-clip videos, so the exclusion slightly favours the frame captioner; at 3 of 300 questions it cannot move an estimate by more than 0.01.
- **Stop rule (C1).** The Claude quota rule stopped Sonnet's secondary arms (question only and 1 frame) after C1's main arms. Sonnet's question-only arm was run later as C8; Sonnet on 1 frame was never run.
- **Token checks (C2, C10).** C2's check expected ≈ 330 Gemini tokens per window and got 280 on split-screen videos; the video arm got the same counts for the same windows, so the rule was restated as "equal to the video arm's tokens" and held for 693 of 761 windows (the rest are short last windows, +2% in total). C10's check omitted a fixed cost of 3 tokens per image block; all three checked calls fit the corrected count exactly.
- **Predictions by type (C3).** Amendment 1 added by-type predictions for C3, after the exploratory by-type analysis of C1 and C2 and before any C3 call.

Deviations.

- Haiku did not keep to the 45-word limit: on the 761 confirmatory windows its captions have a median of 45 words and 47% are over it, against 16% for Sonnet, 10% for Qwen and none for Gemini. The most useful captions (Gemini's) are the shortest, so length does not explain the captioner effect.
- The development entries K1 and K2 were written before their runs but committed after them.
- The open reader (C12) differs from the Claude readers by design: greedy decoding without retry (an unparsed answer counts as wrong), Qwen's default system prompt, the content blocks joined into one message, no thinking, and no cost account.
- The rule for the development step K1 did not say what to do when its second stage was also ambiguous; it was, and the index direction was continued without claiming a tie.

Hand labels. The labels were blind to the judge, but made in a session where Claude (Opus) pointed out caption details on 6 items; κ is 0.81 with the first-instinct labels too.

D Tables by aspect and by video type

Table 5 gives the accuracy of every arm by video type and by aspect. Every stacked question is about speed, and every two-clip question is about order, so the two breakdowns overlap.

Table 6 gives the main differences by video type, with 95% CIs over clip clusters. They are descriptive: with 41–137 questions per type, only differences of about 0.1 or more can show.

Term	Pred.	Result [95% CI]
C1, pre-registered; P1–P3 with 98.33% CIs		
P1 aspect contrast, S reads	−0.20	−0.101 [−0.231, +0.028]
P2 modality, H	−0.08	+0.030 [−0.034, +0.094]
P3 H: Gem. frame − self capt.	+0.06	+0.148 [+0.088, +0.209]
S1 S: Gem. video capt. − frames	−0.07	−0.007 [−0.054, +0.040]
S2 H: Gem. video − frame capt.	+0.02	−0.054 [−0.098, −0.013]
S3 H: Gem. video capt. − frames	0.00	+0.125 [+0.071, +0.178]
S4 aspect contrast, H reads	−0.12	−0.055 [−0.172, +0.060]
S5 H: Gem. video capt. − 1 frame	+0.05	+0.155 [+0.098, +0.212]
S6 H on Gem. video − S on frames	−0.13	−0.061 [−0.111, −0.010]
C2, frame-clip captions (H reads)		
C2a frame-clip − self capt.	+0.10	+0.081 [+0.030, +0.131]
C2b frame − frame-clip capt.	+0.05	+0.067 [+0.030, +0.108]
C2c video − frame-clip capt.	0.00	+0.013 [−0.030, +0.061]
C3, Sonnet captions (H reads)		
C3a Sonnet − self capt.	+0.09	+0.064 [+0.017, +0.111]
C3b Sonnet capt. − frames	+0.12	+0.094 [+0.044, +0.148]
C3c H on Sonnet capt. − S on frames	−0.07	−0.091 [−0.145, −0.040]
C3d Sonnet − Gem. frame-clip capt.	0.00	−0.017 [−0.071, +0.037]
C4, minimal pairs (pair accuracy, cl. CIs)		
C4a H: self capt. − frames	+0.06	+0.093 [+0.028, +0.161]
C4b H: Sonnet − self capt.	+0.09	+0.104 [+0.017, +0.190]
C4c H on Sonnet capt. − S on frames	−0.12	−0.176 [−0.276, −0.079]
C4d H: Gem. video capt. − frames	+0.18	+0.249 [+0.173, +0.319]
C4e S: own capt. − frames	−0.08	−0.176 [−0.260, −0.093]
C5, describe, then answer (H, cl. CIs)		
C5a describe − frames	+0.03	+0.071 [+0.029, +0.114]
C5b describe − self capt.	0.00	+0.040 [−0.011, +0.092]
C6, 448 px frames (H reads)		
C6a 448 − 224 px frames	+0.04	+0.037 [−0.010, +0.084]
C6b Sonnet capt. − 448 px frames	+0.07	+0.057 [+0.003, +0.111]
C6c Gem. video capt. − 448 px frames	+0.08	+0.088 [+0.037, +0.138]
C6d Gem. frame capt. − 448 px frames	+0.13	+0.141 [+0.088, +0.195]
A6, question-blind scores (AUC, cl. CIs)		
H1 motion, stratified (97.5%)	0.60	0.424 [0.281, 0.567]
H2 coverage, stratified (97.5%)	0.58	0.525 [0.380, 0.669]
aspect + reader: + scores	+0.04	+0.034 [+0.011, +0.059]
+ video type: + scores	+0.03	+0.013 [−0.005, +0.030]
A8, frozen decision rule, 738 new questions (H, cl. CIs)		
H1 rule's AUC (97.5%)	0.68	0.666 [0.587, 0.743]
H2 speed shift (97.5%)	−0.25	−0.148 [−0.320, +0.007]
self capt. − frames	+0.03	−0.064 [−0.114, −0.014]
Sonnet capt. − frames	+0.09	+0.008 [−0.026, +0.042]
Gem. video capt. − frames	+0.12	+0.058 [+0.017, +0.098]
rule − always index	+0.005	+0.013 [+0.002, +0.026]
rule − always frames	+0.08	+0.014 [−0.013, +0.041]
speed only − always index	+0.005	+0.013 [−0.002, +0.028]
A9, projected costs on 36 EgoSchema windows (measured / projected − 1)		
cost per question: frames / capt.	±15% / ±25%	+5–6% / +4–11%
Q_{BE}, three downgrade pairs	±30%	−5% to 0%
A10, A8 − C1 on direction, H on frames (cl. CIs)		
A10b standardized by k options	+0.18	+0.261 [+0.129, +0.385]
A10c chance-corrected	+0.25	+0.361 [+0.189, +0.533]
A10d same video, same k	+0.15	+0.202 [+0.083, +0.324]
A10e all aspects, by aspect and k	+0.06	+0.083 [+0.039, +0.128]

Term	Pred.	Result [95% CI]
C7, audit: reader error / misstatement / omission		
H, self capt.	.20 / .30 / .50	.08 / .28 / .63
H, Gem. video capt.	.25 / .30 / .45	.10 / .21 / .69
S, Gem. video capt.	.15 / .35 / .50	.05 / .24 / .71
stacked: omission	$\geq$ .70	.93–1.00
reversed: misstatement	$\geq$ .50	.28–.45
C8, Sonnet on the question only		
C8a S − H, question only	+0.02	+0.003 [−0.061, +0.064]
C8b S: frames − question only	+0.37	+0.384 [+0.320, +0.451]
retention, S: Gem. video / own capt.	0.98 / 0.76	0.98 / 0.77
C9, Sonnet captions of 448 px frames		
C9a H: 448 − 224 px capt.	+0.03	+0.003 [−0.037, +0.047]
C9b H on 448 px capt. − S on frames	−0.06	−0.088 [−0.138, −0.037]
C9c S: own 448 px capt. − frames	−0.07	−0.067 [−0.114, −0.024]
C9d H: 448 px − Gem. frame capt.	−0.06	−0.081 [−0.131, −0.030]
C9e S: own capt., 448 − 224 px	+0.02	+0.020 [−0.024, +0.064]
C10, one caption per video		
C10a H: whole − window capt.	0.00	+0.013 [−0.027, +0.057]
C10b S: whole − window capt.	+0.01	+0.010 [−0.030, +0.051]
C10c H on whole − S on frames	−0.09	−0.077 [−0.128, −0.027]
C10d S: own whole − frames	−0.08	−0.077 [−0.125, −0.034]
C11, aspect-aware captions		
C11a H: aspect-aware − whole	+0.03	+0.027 [−0.013, +0.067]
C11b S: aspect-aware − whole	+0.03	+0.040 [+0.003, +0.081]
C11c H on aspect-aware − S on frames	−0.05	−0.051 [−0.104, +0.000]
C11d S: own aspect-aware − frames	−0.05	−0.037 [−0.081, +0.007]
C12, open-weights reader (Q reads)		
C12a own capt. − frames	−0.01	−0.121 [−0.172, −0.071]
C12b Sonnet − own capt.	+0.09	+0.125 [+0.074, +0.175]
C12c Haiku − own capt.	+0.05	+0.067 [+0.020, +0.114]
C12d Gem. frame-clip − own capt.	+0.08	+0.152 [+0.098, +0.205]
C12e Gem. frame − own capt.	+0.14	+0.178 [+0.125, +0.236]
C12f Gem. video capt. − frames	+0.12	+0.040 [−0.010, +0.091]
C12g Q − H, on Sonnet capt.	−0.05	−0.047 [−0.094, +0.000]
C12h Q − H, question only	0.00	−0.013 [−0.077, +0.051]
C13, question-aware captions, read without the frames		
C13a H: question-aware − blind self capt.	+0.04	+0.017 [−0.034, +0.067]
C13b S: question-aware − blind own capt.	+0.05	+0.017 [−0.030, +0.067]
C13c S: own question-aware capt. − frames	−0.04	−0.071 [−0.114, −0.027]
C13d H on Sonnet q.-aware − S on frames	−0.05	−0.067 [−0.111, −0.024]
C13e H: q.-aware self capt. − describe	0.00	−0.024 [−0.074, +0.024]
C13f H: Sonnet q.-aware − blind capt.	+0.04	+0.024 [−0.024, +0.071]
C14, Sonnet on others' captions; a re-read		
C14a S − H, on Haiku's self capt.	+0.03	+0.034 [−0.010, +0.081]
C14b S − H, on Gem. frame capt.	+0.05	+0.020 [−0.013, +0.054]
C14c S − H, on Gem. frame-clip capt.	+0.05	+0.064 [+0.027, +0.101]
C14d own-words interaction	−0.03	−0.030 [−0.088, +0.027]
C14e S: Sonnet − Haiku capt.	+0.04	+0.034 [−0.017, +0.081]
C14f H: re-read − first read	0.00	−0.013 [−0.047, +0.020]
C14g S: Gem. frame capt. − frames	+0.04	+0.013 [−0.034, +0.061]

Table 4: Every term with a prediction written before its run, in the order run within each column. The labels (P = pre-registered, S = secondary terms of C1, C2–C14 = follow-ups, A6 and A8 = the decision-rule entries, Appendix G, A9 and A10 = checks of the cost projection and of A8's shift on frames) are the research log's. H = Haiku, S = Sonnet, Q = Qwen reading; capt. = captions. CIs are paired bootstrap over questions unless marked cl. (clip clusters).

E Main results in detail

This appendix gives the parts of §4 that the main text shortens, in its order. Table 7 collects the four effects of §3.

The captioner effect (§4.1). With Qwen reading, the captioner effect has the clip-cluster CI [+0.070, +0.182]. On the further questions about the same videos, Gemini's video captions beat Haiku's self-captions by +0.122, cl. [+0.068, +0.177] (computed after that run, not predicted; Appendix G).

Readers on other indexes (§4.2). Figure 2 shows accuracy by reader. On frames, Haiku and Qwen are both below Sonnet (Sonnet − Qwen +0.141 [+0.084, +0.199]). On Gemini's three indexes Sonnet keeps 0.97–1.04 of its frames' gain over the question alone, against 0.68, cl. [0.56, 0.81], on Haiku's self-captions (retention, §3). Qwen's gap to Haiku has a CI that excludes 0 only on Gemini frame captions. We predicted Qwen's gap on Sonnet's captions (−0.05) and Sonnet's gaps on Haiku's captions and on Gemini's frame and frame-clip captions (Table 4). The other reader gaps, including the tie on Sonnet's four indexes, were not predicted. Every gap with its CI, and the retention of every index, are in Appendix F.

Reader, input	plain (137)	two-clip (45)	reversed (74)	stacked (41)	action (60)	attr. (60)	direction (60)	order (57)	speed (60)	all (297)
Haiku, question only	0.60	0.31	0.16	0.34	0.63	0.37	0.28	0.37	0.40	0.41
Haiku, 1 frame	0.77	0.62	0.27	0.49	0.88	0.55	0.32	0.65	0.52	0.58
Haiku, 5 frames	0.80	0.60	0.38	0.41	0.93	0.67	0.28	0.65	0.53	0.61
Haiku, 5 frames, 448 px	0.82	0.62	0.43	0.49	0.98	0.68	0.38	0.65	0.55	0.65
Haiku, describe, then answer	0.82	0.69	0.50	0.54	0.90	0.73	0.47	0.72	0.60	0.68
Haiku, self-captions	0.80	0.82	0.43	0.32	0.90	0.70	0.38	0.82	0.42	0.64
Haiku, Sonnet captions	0.85	0.93	0.45	0.44	0.97	0.72	0.42	0.91	0.53	0.71
Haiku, Sonnet captions, 448 px	0.88	0.87	0.46	0.41	0.95	0.77	0.43	0.84	0.57	0.71
Haiku, Sonnet whole-video	0.85	0.89	0.57	0.37	0.97	0.83	0.43	0.89	0.48	0.72
Haiku, Sonnet aspect-aware	0.88	0.96	0.58	0.39	0.95	0.88	0.43	0.95	0.53	0.75
Haiku, Gemini frame-clip	0.88	0.87	0.57	0.34	0.98	0.85	0.43	0.88	0.48	0.72
Haiku, Gemini frame	0.92	1.00	0.64	0.41	1.00	0.90	0.55	1.00	0.52	0.79
Haiku, Gemini video	0.86	0.93	0.53	0.49	0.95	0.82	0.43	0.93	0.57	0.74
Sonnet, question only	0.61	0.29	0.18	0.32	0.70	0.38	0.23	0.40	0.35	0.41
Sonnet, 5 frames	0.89	0.91	0.64	0.66	0.97	0.82	0.58	0.91	0.72	0.80
Sonnet, Sonnet captions	0.88	0.91	0.46	0.39	0.97	0.73	0.43	0.89	0.53	0.71
Sonnet, Sonnet captions, 448 px	0.88	0.96	0.49	0.41	0.95	0.77	0.48	0.95	0.52	0.73
Sonnet, Sonnet whole-video	0.90	0.93	0.51	0.27	0.98	0.82	0.45	0.91	0.45	0.72
Sonnet, Sonnet aspect-aware	0.92	0.96	0.55	0.39	1.00	0.87	0.45	0.95	0.55	0.76
Sonnet, Gemini video	0.93	0.98	0.57	0.54	1.00	0.82	0.55	0.98	0.62	0.79
Qwen, question only	0.50	0.42	0.23	0.32	0.58	0.38	0.28	0.44	0.30	0.40
Qwen, 5 frames	0.86	0.58	0.45	0.44	0.93	0.75	0.38	0.67	0.55	0.66
Qwen, Haiku captions	0.81	0.58	0.45	0.22	0.90	0.70	0.42	0.63	0.37	0.60
Qwen, Sonnet captions	0.86	0.64	0.51	0.27	0.98	0.78	0.43	0.70	0.40	0.66
Qwen, Gemini frame-clip	0.88	0.67	0.54	0.32	0.95	0.85	0.45	0.72	0.47	0.69
Qwen, Gemini frame	0.91	0.71	0.58	0.32	0.98	0.87	0.50	0.75	0.47	0.71
Qwen, Gemini video	0.89	0.76	0.50	0.34	0.97	0.80	0.47	0.77	0.48	0.70
Qwen, Qwen captions	0.75	0.60	0.26	0.24	0.90	0.53	0.28	0.58	0.38	0.54

Table 5: Accuracy by video type and by aspect, every arm (297 confirmatory questions; n in brackets).

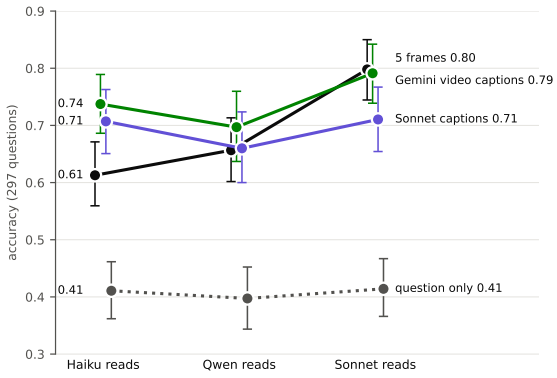


Figure 2: Accuracy by reader, in order of accuracy on frames, for the question alone, 5 frames, and the two indexes every reader read. 95% CIs over clip clusters. The frames line crosses both index lines; Sonnet's captions are flat from Haiku to Sonnet, Gemini's are not.

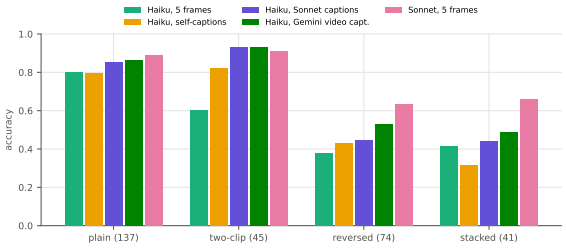


Figure 3: Accuracy by video type for five arms (297 questions; n per type in brackets).

Video types (§4.3). Figure 3 gives accuracy by video type. On reversed videos, Haiku on Sonnet's self-captions trails Sonnet on frames by −0.19 (Table 6). There Haiku is no more accurate on Sonnet's captions than on its self-captions (+0.01 [−0.08, +0.10]), although we predicted a gain, and it is more accurate on Gemini's frame-clip captions (+0.14 [+0.04, +0.24]). A reader that misses the reversal falls back to the prior. Of Sonnet's wrong answers on reversed videos, 0.59 (frames) and 0.65 (self-captions) equal its question-only answer, against 0.43 if errors were uniform. On stacked videos the captions do not say how the copies differ, so readers give the same answer on both videos of a pair more often than Sonnet on frames does (Appendix H). On two-clip videos, describing each window before answering gains only +0.09 [−0.06, +0.24].

Perception and omission by video type. A low caption score could mean that the captioner did not perceive the fact, or that it perceived the fact and did not write it down. Sonnet also reads the same pixels with the question in view, so for Sonnet the arms separate the two. Table 8 (descriptive) splits Sonnet's headroom into a perception shortfall, 1 − frames, and the modality effect, self-captions − frames, which is what Sonnet sees when asked but does not write down blind. On plain and two-clip videos the modality effect is ≈ 0. On reversed videos Sonnet is misled even with the question

Difference	plain (137)	two-clip (45)	reversed (74)	stacked (41)
P2 modality, H: self capt. − frames	−0.01 [−0.07, +0.06]	+0.22 [+0.10, +0.35]	+0.05 [−0.04, +0.15]	−0.10 [−0.26, +0.05]
P3 H: Gem. frame − self capt.	+0.12 [+0.06, +0.19]	+0.18 [+0.09, +0.28]	+0.20 [+0.09, +0.32]	+0.10 [+0.02, +0.19]
C2a H: Gem. frame-clip − self capt.	+0.08 [+0.01, +0.15]	+0.04 [−0.09, +0.17]	+0.14 [+0.04, +0.24]	+0.02 [−0.08, +0.14]
C3a H: Sonnet − self capt.	+0.06 [−0.01, +0.12]	+0.11 [−0.03, +0.24]	+0.01 [−0.08, +0.10]	+0.12 [−0.03, +0.28]
C3b H: Sonnet capt. − frames	+0.05 [+0.00, +0.11]	+0.33 [+0.20, +0.48]	+0.07 [−0.04, +0.18]	+0.02 [−0.16, +0.21]
C3c H on Sonnet capt. − S on frames	−0.04 [−0.10, +0.02]	+0.02 [−0.10, +0.13]	−0.19 [−0.32, −0.07]	−0.22 [−0.44, +0.00]
S1 S: Gem. video capt. − frames	+0.04 [−0.01, +0.09]	+0.07 [−0.02, +0.16]	−0.07 [−0.19, +0.05]	−0.12 [−0.31, +0.05]
S: own capt. − frames	−0.01 [−0.06, +0.03]	+0.00 [−0.06, +0.07]	−0.18 [−0.30, −0.05]	−0.27 [−0.47, −0.09]
S3 H: Gem. video capt. − frames	+0.06 [−0.01, +0.12]	+0.33 [+0.22, +0.47]	+0.15 [+0.04, +0.27]	+0.07 [−0.03, +0.17]
C5a H: describe − frames	+0.02 [−0.04, +0.08]	+0.09 [−0.06, +0.24]	+0.12 [+0.01, +0.23]	+0.12 [+0.00, +0.23]
C5b H: describe − self capt.	+0.03 [−0.04, +0.10]	−0.13 [−0.30, +0.02]	+0.07 [−0.04, +0.18]	+0.22 [+0.09, +0.37]
C10a H: whole − window capt.	+0.00 [−0.04, +0.04]	−0.04 [−0.14, +0.07]	+0.12 [+0.03, +0.23]	−0.07 [−0.21, +0.06]
C10b S: whole − window capt.	+0.02 [−0.01, +0.06]	+0.02 [−0.05, +0.10]	+0.05 [−0.06, +0.16]	−0.12 [−0.29, +0.05]
C11a H: aspect-aware − whole	+0.02 [−0.03, +0.07]	+0.07 [−0.03, +0.15]	+0.01 [−0.08, +0.11]	+0.02 [−0.08, +0.12]
C11b S: aspect-aware − whole	+0.02 [−0.02, +0.07]	+0.02 [−0.05, +0.09]	+0.04 [−0.05, +0.14]	+0.12 [+0.00, +0.26]
C12a Q: own capt. − frames	−0.11 [−0.18, −0.04]	+0.02 [−0.10, +0.13]	−0.19 [−0.31, −0.08]	−0.20 [−0.34, −0.05]
C12b Q: Sonnet − own capt.	+0.11 [+0.05, +0.18]	+0.04 [−0.09, +0.19]	+0.26 [+0.13, +0.39]	+0.02 [−0.10, +0.16]
C12g Q − H, on Sonnet capt.	+0.01 [−0.04, +0.05]	−0.29 [−0.40, −0.17]	+0.07 [−0.03, +0.17]	−0.17 [−0.28, −0.05]

Table 6: Differences in accuracy by video type (95% CIs over clip clusters). H = Haiku, S = Sonnet, Q = Qwen reading; term labels as in Table 4.

Effect	Arms compared	Claude chain	Gemini chain
Source	video captions − frame-clip captions	0 by design	+0.013 [−0.030, +0.061]
Visual budget	frame captions − frame-clip captions	0 by design (+0.003 at 448 px)	+0.067 [+0.030, +0.108]
Captioner	other captioner − self-captions, same pixels	+0.064 [+0.017, +0.111]	+0.081 [+0.030, +0.131]
Modality	self-captions − 5 frames	+0.030 [−0.034, +0.094]	(same)
Total	captions − 5 frames	+0.094 [+0.044, +0.148]	video: +0.125; frame: +0.178

Table 7: The four effects, with Haiku reading. Accuracy differences on 297 questions, 95% paired bootstrap CIs (the modality cell is the pre-registered P2, with its 98.33% CI). In the Gemini chain the terms add up along two paths (§3).

Type (n)	Q. only	Frames	Self-capt.	Modality	Retention
plain (137)	0.61	0.89	0.88	−0.01 [−0.06, +0.03]	0.95 [0.79, 1.12]
two-clip (45)	0.29	0.91	0.91	+0.00 [−0.06, +0.07]	1.00 [0.91, 1.12]
reversed (74)	0.18	0.64	0.46	**−0.18** [−0.30, −0.05]	0.62 [0.35, 0.88]
stacked (41)	0.32	0.66	0.39	**−0.27** [−0.47, −0.09]	0.21 [−0.27, 0.67]
all (297)	0.41	0.80	0.71	−0.09 [−0.14, −0.04]	0.77 [0.64, 0.90]

Table 8: Where Sonnet's headroom goes, by video type (Sonnet as captioner and reader; 95% CIs over clip clusters). Modality = self-captions − frames; the perception shortfall is 1 − frames.

Reader arm	Reader err.	Misstat.	Omission	n
H, self-captions	0.08	0.28	0.63	106
H, Sonnet capt.	0.13	0.26	0.61	87
H, Gem. frame-clip	0.17	0.15	0.68	82
H, Gem. frame	0.10	0.18	0.73	62
H, Gem. video	0.10	0.21	0.69	78
S, Gem. video	0.05	0.24	0.71	62
S, Sonnet capt.	0.08	0.26	0.66	86

Table 9: Caption audit (§4.4): share of each reader arm's wrong answers by the judge's label of the captions. Reader error = the captions state the correct option; misstatement = a wrong option; omission = neither. n = wrong answers.

in view (shortfall 0.36), and writing blind loses another 0.18. On stacked videos it sees the comparison when asked (0.66), but its blind captions keep little of it (retention 0.21, wide CI). For Haiku reading Sonnet's captions, captions − frames is positive on every type, with a CI excluding 0 only on two-clip videos (Table 12).

The audit (§4.4). Two readers share some caption sets. Built videos hold 0.70–0.82 of each arm's omissions. Misstatements are 0.15–0.28 of the wrong answers, and reader errors are below the predicted 0.15–0.25. As predicted, 0.93–1.00 of stacked failures are omissions in every arm (predicted ≥ 0.70). On reversed videos, misstatements (forward narration) are only 0.28–0.45 (predicted ≥ 0.50) and omissions 0.45–0.62, so the captions either narrate the forward event or do not commit to a direction. The judge also wrote one of the caption sets it labels. The hand-labelled items were stratified by the judge's label, and the threshold for κ, 0.6, was set in advance. All 5 disagreements are items the author left "silent", so if anything Table 9 undercounts omission. The audit by video type is in Appendix F.

Question-aware arms (§4.5). Describing before answering was predicted to gain +0.03 over the frames. Its gain is on reversed and stacked videos

(+0.12 each), and it costs 2× the frames arm per question ($0.000415 vs. $0.000204), mostly in thinking tokens. The question-aware captions' gains over the blind captions have the CIs [−0.034, +0.067] (Haiku) and [−0.030, +0.067] (Sonnet), and Sonnet on them was predicted to trail its frames by −0.04. Haiku on Sonnet's question-aware captions trails Sonnet on frames by −0.067 [−0.111, −0.024] (on the blind ones, −0.091). On reversed videos, Sonnet's gap to its frames on its question-aware captions has the CI [−0.26, −0.04]. Splitting the describe-then-answer call into a caption and a separate read loses nothing measurable (−0.024 [−0.074, +0.024]; predicted 0.00). The question-aware captions keep the 45-word limit (median 43–45 words), and 1.1% (Haiku) and 0.1% (Sonnet) name an option letter. The aspect-aware gain with Sonnet reading has cl. [0.000, +0.081], and both aspect-aware gains were predicted at +0.03. Results by type are under *Question-aware captions by type* and *Task-aware captions* in Appendix F.

Reader, input	Acc.	In tok.	$/q	Index
H, 1 frame	0.582	1,189	1.46	—
H, 5 frames 448 px	0.650	2,676	3.39	—
H, describe, answer	0.684	1,579	4.15	—
H, Sonnet 448 px	0.710	1,279	1.80	11.67
H, Sonnet whole-video	0.721	1,194	1.56	4.92
H, Sonnet aspect-aware	0.747	1,213	1.62	5.27
S, Sonnet 448 px	0.731	1,281	31.03	11.67
S, Sonnet whole-video	0.721	1,196	28.49	4.92
S, Sonnet aspect-aware	0.761	1,215	28.97	5.27
H, self-capt., re-read	0.630	1,286	1.89	0.58
H, self-capt., question-aware	0.660	1,298	1.73	1.03
H, Sonnet question-aware	0.731	1,296	1.63	9.74
S, Haiku capt.	0.677	1,265	31.62	0.58
S, Gem. frame-clip	0.788	1,214	28.74	7.60
S, Gem. frame	0.811	1,211	28.71	14.61
S, Sonnet question-aware	0.727	1,275	29.89	9.74
Q, question only	0.397	121	—	—
Q, 5 frames	0.657	529	—	—
Q, Qwen capt.	0.535	268	—	—
Q, Haiku capt.	0.603	301	—	0.58
Q, Sonnet capt.	0.660	295	—	9.10
Q, Gem. frame-clip	0.687	263	—	7.60
Q, Gem. frame	0.714	261	—	14.61
Q, Gem. video	0.697	264	—	7.26

Table 10: The variant arms, the arms added by C13–C14, and Qwen's eight arms (Q = Qwen reading), as in Table 1. Qwen runs on a free GPU, so it has no price; its input tokens have no CLI system prompt. The question-aware captions are written for the one question asked of each video, so their Index cost is spent per question, not once per video.

F Additional results

Visual budget within Claude. Sonnet's captions of 448 px frames (up to 720 visual tokens per window, 3.6× as many) are no more useful to Haiku than its captions of the 224 px frames: +0.003 [−0.037, +0.047], against a predicted +0.03. With power ≈ 0.25 for +0.03, this is weak evidence against a small effect, but the CI rules out more than about +0.045. The two budget changes also differ in size (200 → 720 tokens here, ≈ 300 → 5,500 for Gemini).

Haiku on 448 px frames. At 448 px, Haiku's frames take 2,676 input tokens per question, 1.76× the 224 px arm, and gain +0.037 [−0.010, +0.084]. Captions at ≈ 1,280 input tokens still beat them: Gemini frame captions by +0.141 [+0.088, +0.195], Gemini video captions by +0.088 [+0.037, +0.138], and Sonnet captions by +0.057 [+0.003, +0.111]. The advantage sits on two-clip videos (+0.31 to +0.38), and for Gemini frame captions also on plain and reversed ones. On stacked videos, every caption arm is within ±0.07 of the 448 px frames.

Task-aware captions. The aspect-aware bullet names the kinds of question, but not how the benchmark builds its videos. It halves Sonnet's gap to its own frames (−0.077 → −0.037 [−0.081, +0.007]), but not on stacked videos, where Sonnet on frames stays 0.27 ahead, and mentions of reversal do not rise. The captions are also 15% longer, so part of the gain may be more words rather than the right ones.

Window length. We checked the per-window design against one Sonnet caption per video, from the same frames, with the length limit summed over the windows. Whole-video captions are no better: +0.013 [−0.027, +0.057] with Haiku reading, and +0.010 [−0.030, +0.051] with Sonnet (predicted +0.00 and +0.01). Splitting into 5 s windows thus costs at most ≈ 0.05–0.06, and helps by at most ≈ 0.03. On one-window videos, where the two captioning calls are identical, the difference is ≈ 0 for both readers (+0.02 and −0.02), as it should be. We did not predict one by-type pattern. Haiku gains +0.12 [+0.03, +0.23] on reversed videos from whole-video captions, while stacked videos lean the other way for both readers (−0.07 and −0.12; CIs include 0).

Resampling noise. On the 48 one-window videos, the captioning call is identical in both arms.

Index	Q_{BE}	Acc. difference
Gemini video	1.8 [1.7, 2.0]	−0.061 [−0.111, −0.010]
Gemini frame-clip	1.9 [1.8, 2.1]	−0.074 [−0.125, −0.024]
Sonnet captions	2.3 [2.2, 2.4]	−0.091 [−0.145, −0.040]
Sonnet, 448 px	3.0 [2.8, 3.1]	−0.088 [−0.138, −0.037]
Sonnet whole-video	1.2 [1.2, 1.3]	−0.077 [−0.128, −0.027]
Sonnet aspect-aware	1.3 [1.3, 1.4]	−0.051 [−0.104, +0.000]
Gemini frame	3.7 [3.5, 4.0]	−0.007 [−0.057, +0.044]

Table 11: Reader downgrade: Haiku on an index against Sonnet on 5 frames. Q_{BE} is the number of questions per video after which the index costs less in total (95% CI over questions).

None of the 48 caption pairs is identical, and the reader's answer changes on 10% (Haiku) to 17% (Sonnet) of those questions, as often as removing the split changes it on videos with two or more windows (17%). Per question, a caption is one sample.

Aspects (P1, null). The pre-registered aspect contrast, that captions lose direction and speed but keep the discrete aspects, did not pass: −0.101 [−0.231, +0.028] for Sonnet, half the predicted −0.20. Speed is the only aspect where captions lose consistently (−0.10 for Sonnet). Aspect nearly coincides with video type (Table 5).

Thinking (not predicted). At effort medium, the model decides per call whether to think, so thinking is observed, not set. On Gemini's captions, thinking marks wrong answers: accuracy is 0.84–0.90 on calls without thinking and 0.62–0.65 on calls with it. For Haiku on frames it does not (0.61 both ways). Sonnet thinks 3.3× less on captions than on frames.

Retention of every index (95% CIs over clip clusters). Haiku: self-captions 1.15 [0.91, 1.46], Sonnet captions 1.47 [1.18, 1.95], Sonnet 448 px 1.48 [1.20, 1.95], Sonnet whole-video 1.53 [1.22, 2.07], Gemini frame-clip 1.55 [1.27, 2.02], Gemini video 1.62 [1.33, 2.10], Gemini frame 1.88 [1.53, 2.51]. Every point estimate exceeds 1, since Haiku is more accurate on every index than on its frames. Sonnet: self-captions 0.77 [0.64, 0.90], 448 px 0.82 [0.71, 0.93], whole-video 0.80 [0.67, 0.93], Gemini video 0.98 [0.85, 1.13]. Qwen: self-captions 0.53 [0.34, 0.70], Sonnet captions 1.01 [0.85, 1.20], Gemini video 1.16 [0.98, 1.37].

Readers on shared indexes. Sonnet − Haiku on the same captions (95% CIs; clip clusters agree): Sonnet captions +0.003 [−0.040, +0.047], Sonnet 448 px +0.020 [−0.024, +0.064], Sonnet whole-video +0.000 [−0.040, +0.040], Sonnet aspect-aware +0.013 [−0.024, +0.051], Gemini video +0.054 [+0.017, +0.091] (plain +0.07 [+0.02, +0.11] over clip clusters; the other types +0.04 to +0.05, CIs include 0). Qwen − Haiku: Haiku captions −0.040 [−0.091, +0.013], Sonnet captions −0.047 [−0.094, +0.000], Gemini frame-clip −0.037 [−0.081, +0.007], Gemini video −0.040 [−0.088, +0.003], Gemini frame −0.077 [−0.125, −0.030]; on two-clip videos −0.18 to −0.29 (every CI excludes 0), on plain ones −0.01 to +0.03. The means were known from Tables 1 and 10 when this analysis was written; its terms and reading rule were logged before the CIs were computed. C14 then added three shared indexes for Sonnet, predicted before its reads (clip-cluster CIs agree): Haiku's self-captions +0.034 [−0.010, +0.081], Gemini frame +0.020 [−0.013, +0.054], and Gemini frame-clip +0.064 [+0.027, +0.101] (two-clip +0.13, stacked +0.17, by clip clusters). The own-words interaction, (Sonnet − Haiku on Sonnet's captions) − (Sonnet − Haiku on Haiku's), is −0.030 [−0.088, +0.027]. For Sonnet, Sonnet's captions beat Haiku's by +0.034 [−0.017, +0.081], half Haiku's captioner effect (+0.064). On stacked videos Gemini's frame captions also lose to Sonnet's frames (−0.20 [−0.35, −0.05]).

Question-aware captions by type. Clip-cluster CIs, descriptive. Sonnet's question-aware captions − its frames: plain −0.05 [−0.10, −0.01], two-clip +0.02 [−0.07, +0.12], reversed −0.15 [−0.26, −0.04], stacked −0.10 [−0.28, +0.05]; the blind captions' are −0.01, 0.00, −0.18 and −0.27. Question-aware − blind captions, by the same captioner and reader: Haiku −0.01, 0.00, −0.04, +0.22 [+0.07, +0.39]; Sonnet −0.04, +0.02, +0.03, +0.17 [−0.03, +0.38]. Against describing before answering, Haiku's question-aware captions gain on two-clip videos (+0.13 [+0.05, +0.23]) and lose on reversed ones (−0.11 [−0.21, −0.01]). Sonnet's retention rises from 0.77 [0.64, 0.90] to 0.82 [0.70, 0.92]. Captioning with the question cost Haiku 1.8× its blind captions per video ($1.03 against $0.58 $\times 10^{-3}$) and Sonnet 1.07× (Table 10).

The open reader in detail. Qwen reads with greedy decoding and the same prompts. Although it is not separated from Haiku on frames, Haiku's captions are +0.067 [+0.020, +0.114] more useful to it than its own. Qwen reads Sonnet's captions 0.047 below Haiku ([−0.094,

Type (n)	Question only	Frames	Sonnet captions	Perception shortfall	Captions − frames
plain (137)	0.60	0.80	0.85	0.20 [0.13, 0.26]	+0.05 [−0.00, +0.11]
two-clip (45)	0.31	0.60	0.93	0.40 [0.30, 0.52]	+0.33 [+0.20, +0.48]
reversed (74)	0.16	0.38	0.45	0.62 [0.51, 0.73]	+0.07 [−0.04, +0.18]
stacked (41)	0.34	0.41	0.44	0.59 [0.45, 0.71]	+0.02 [−0.16, +0.21]
all (297)	0.41	0.61	0.71	0.39 [0.33, 0.44]	+0.09 [+0.04, +0.15]

Table 12: Table 8 for Haiku reading Sonnet's captions (95% CIs over clip clusters). Retention is 1.47 [1.18, 1.95] overall; by type it is not reported, because where Haiku's frames barely beat the question alone (stacked) its CI is unbounded.

+0.000]; predicted −0.05): level on plain (+0.01) and reversed (+0.07) videos, and below on two-clip (−0.29 [−0.40, −0.17]) and stacked (−0.17 [−0.28, −0.05]) videos, where the answer must be combined across windows or copies. On the question alone it does not differ measurably from Haiku (−0.013, cl. [−0.076, +0.047]). Its self-captions lose most on reversed (0.26) and stacked (0.24) videos. The 49 unparsed answers are refusals, 42 of them on caption arms. Without them, the captioner effect is +0.106, cl. [+0.052, +0.164], and the modality effect −0.103, cl. [−0.152, −0.055]. Its captions are shorter than Claude's (median 30 words; 10% over the 45-word limit).

Audit by video type. Two-clip failures are few (1–8 per arm) and mostly reader errors, where the captions state the order and the reader still answers wrongly. Plain failures are 0.60–0.80 omission; stacked failures 0.93–1.00. The hard items fall into four kinds: a static description that fits a wrong option; a near-synonym of an option; a 1 s last window that contradicts the long first one; and an answer that needs knowledge of the task (debris "tumbling upward" means "rebuilding" only to a reader who knows that clips may be reversed).

More on cost. Figure 4 plots accuracy against cost per question. Captions cut the video-content tokens to 43% of the frames', but the input tokens only to 81%. Sonnet on its self-captions repays after $Q_{\mathrm{BE}} = 8.9$ [7.8, 10.2], at −0.088. Haiku on Sonnet captions is +0.057 more accurate than Haiku on 448 px frames and cheaper per question, and repays after 57.5 [53.3, 62.1]. Latency does not favour captions: the median call on captions is 0.1–1.1 s slower than on frames with the same reader (CLI wall clock with start-up, not controlled for time of day). Thinking is 2–5% of reader tokens.

Longer videos. The clips average 2.6 windows, so the fixed reader prompt is a large share of each question's cost. We project the measured costs by Proposition 2: each arm's cost per question is its mean plus its input tokens per window (exact by construction: 200 for frames, 63–87 for captions) at its input price per extra window, and the index costs its mean per window times W. This assumes that output and thinking tokens do not grow with W, and that the reader reads every window. For one reader, the saving per question grows with length: Gemini video captions cost Haiku 0.82 of the frames' price per question at 2.6 windows, 0.59 at one minute (12 windows) and 0.38 at ten minutes (120), and Sonnet 0.70, 0.53 and 0.37. The break-even does not fall, because captioning grows with every window too: Haiku's Q_{BE} stays at 203–214, and Sonnet's rises from 5.9 to a limit of 10.7 (at the measured 2.6 windows the projection gives 203 and 5.9, against the measured 202 and 5.8), since its fixed saving (less thinking on captions) counts for less on long videos. For the reader downgrade, the per-window indexes' Q_{BE} of 1.8–3.7 rises to 4.5–8.9 at one minute and to a limit of 7.2–14.5, while the cost per question falls from 1/23–1/25 to 1/50 of Sonnet on frames. The limit does not depend on the reader's fixed prompt, including the CLI's. A reader that retrieves a few windows instead of reading all saves less per question, so for it these Q_{BE} are lower bounds.

The projection against measured costs. The EgoSchema development run (39 videos of 36 windows; Limitations) read every window with the same arms and prompts, so it measures the costs that Proposition 2 projects; contamination does not bear on costs. Its frames are 224×168, so each takes 48 image tokens, not 40. Projected from the TempCompass measurements with that correction (8 tokens per frame, for reads and for Claude's captioning), every arm's cost per question is within 11% of the measured one (measured / projected 1.05–1.06 for frame arms, 1.04–1.11 for caption

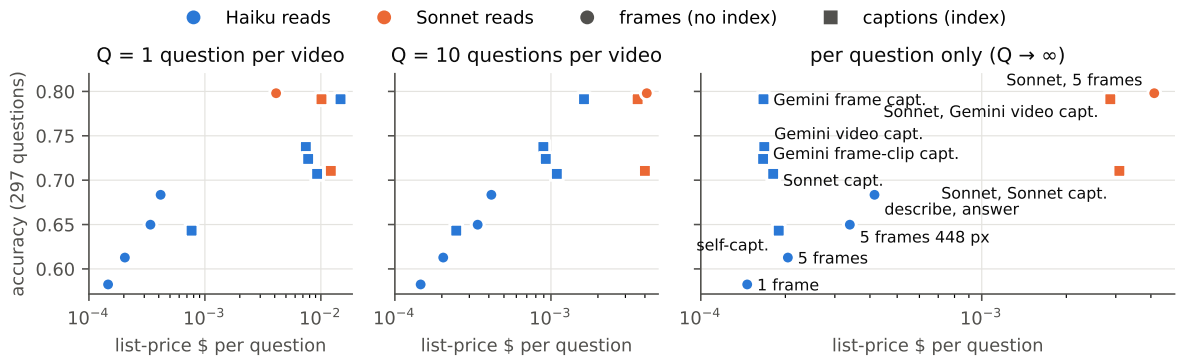


Figure 4: Accuracy against list-price $ per question (log scale), one point per arm, with the index cost spread over Q = 1, 10 and infinitely many questions per video (§5). Circles are frame arms, and squares are index arms. Question only (0.41 for both readers) is below the axis; the 448 px caption, whole-video and aspect-aware variants are in Table 10 only.

arms), the index cost per video within 2%, and every Q_{BE} within 7%: 6.3 measured against 6.6 projected for Haiku on Sonnet's captions against Sonnet on frames, 11.3 against 11.4 for Sonnet on its captions, and 223 against 239 for Haiku on Sonnet's captions against Haiku on frames. The reads take 7–10% more input tokens than projected, and output tokens differ by −41% to +57%, but output is under 120 tokens per read. Without the correction, measured Q_{BE} is 0.70–0.84 of projected. The run's Q_{BE} for the first pair, 6.3 [6.1, 6.4], was predicted in writing to fall in [5, 10]; the full comparison was written down before it was computed, after the run (research log A9).

Question-level and clustered CIs. For every pair of the 28 arms on the confirmatory questions with any difference (378), we computed both 95% CIs. The median ratio of their widths (clusters / questions) is 1.00 (IQR 0.95–1.05), since most of the 156 clusters hold one or two questions. In 6 pairs one CI excludes 0 and the other does not, each with an endpoint within 0.003 of 0. Five are comparisons the paper does not report; the sixth is C11b (§4.5), +0.040 [+0.003, +0.081], cl. [0.000, +0.081]. On the 15 pairs of Haiku and Sonnet arms on minimal pairs, the two never differ, and the clustered CIs are wider (median ratio 1.17). This check was run after the paper was drafted (research log D7).

Forced choice over 1 s gaps. In an earlier phase of this project, 160 items asked which candidate fills a masked 1 s clip of a 60 s video. Copying the adjacent frame chose correctly 87.5% of the time, level with Claude Opus reading the surrounding captions (85.0%).

Cost without the CLI's prompt. The smallest Claude question-only call has 1,003 input tokens, while its text (system prompt, reader prompt and question) is ≈ 120–150 tokens, and Qwen reads the same text in 121; so the CLI adds ≈ 850–900 tokens per call. Removing 850 input tokens from every Claude reader and caption call (Gemini unchanged) gives, for the reader downgrade, Q_{BE} 3.1 (Gemini video), 3.3 (frame-clip), 6.3 (frame), 2.0 (Sonnet), 3.2 (Sonnet 448 px), 1.4 (whole-video) and 1.5 (aspect-aware), at 1/25 to 1/34 of Sonnet's cost on frames per question. Sonnet on its self-captions repays after 4.6 instead of 8.9. Removing 700 or 1,000 tokens moves every Q_{BE} by at most 0.8. For one reader with a Gemini index, Q_{BE} is unchanged, since the fixed cost cancels, and the saving per question rises to 30% (Haiku) and 52% (Sonnet). For the reader downgrade, Q_{BE} rises about 1.7× for the Gemini indexes and barely moves for Sonnet's, whose caption calls lose the prompt too.

G The decision rule in detail

Three research-log entries, each committed before it ran: A6 (two scores as predictors, no calls), A7 (development of the rule, no calls) and A8 (the test, 2,952 Haiku calls).

In brief. Haiku's self-captions win 34 of the questions where they and its frames disagree and lose 25, and Qwen on Sonnet's captions wins 26 and loses 25. A6's two scores recover how the videos were built (two-clip videos move most and lose least), and they add nothing to the video type. In

A7, a selection rule fixed in advance chose aspect + pair. Alone, the pair ranks at 0.695 [0.640, 0.749] and the aspect at 0.602 [0.522, 0.673]. Within a pair, the rule ranks at 0.599 [0.510, 0.676] (computed after the run). For decisions, most of the rule's value is the choice per pair. Against the best fixed choice per pair it gains +0.005 [−0.003, +0.014] in accuracy, and its one signal within a pair is that speed questions favour the frames. In A8, we did not predict one shift. Haiku's frames score 0.699 on these questions against 0.613 on the confirmatory ones, mostly on reworded direction questions, and the number of options does not explain it (below). For a weak reader, the value of the index thus depends on the question set. The speed shift is P(index right | speed) − P(index right | other aspects) on disagreements, 0.39 against 0.54, and its sign holds for all three indexes. The rule's gain over always reading the index was predicted at +0.005.

A6: question-blind scores. *Motion* is the mean SigLIP 2 cosine distance between consecutive 224 px frames, 1 s apart, over the whole video. *Coverage* is, per captioner, the mean over windows of the SigLIP 2 cosine between the window's caption (text tower `siglip2-base-patch16-224`, in the frames' space) and the mean of its frames. A gate run before any outcome was joined checked that each caption is closer to its own window than to a random window of another video: it is in every window, for both captioners. A loss is a question that a reader answers right from its 5 frames and wrong from its self-captions; the rows are the 419 (reader, question) pairs with the frames right (Haiku and Sonnet), with 63 losses. The statistic is a stratified AUC: over pairs of a loss and a kept answer from the same video type and reader, the share where the loss has the higher motion (lower coverage). By reader, motion gives 0.45 (Haiku) and 0.40 (Sonnet), coverage 0.45 and 0.59; on plain videos, 0.45 and 0.51 (17 losses); for gains instead of losses, 0.52 and 0.42; no CI excludes 0.5. By aspect (descriptive), motion cuts both ways: more motion means more losses on speed questions (0.64 [0.45, 0.78]) and fewer on direction questions (0.27 [0.11, 0.47]), where a slow reversed clip gives the caption little to commit a direction to. As classifiers (out-of-fold logistic regression, 10 folds by clip cluster), aspect + reader gives AUC 0.751, + motion and coverage 0.785, + video type instead 0.782, and both 0.794: the scores add what the video type adds, and nothing on top of it.

Model	Clusters held out	Pair held out too
M1 aspect + pair	**0.756** [0.705, 0.803]	0.715 [0.659, 0.763]
M2 M1 + video	0.758 [0.706, 0.807]	0.727 [0.670, 0.779]
M3 M1 + captions	0.751 [0.700, 0.799]	0.711 [0.653, 0.759]
M4 M1 + video + captions	0.754 [0.702, 0.802]	0.721 [0.664, 0.772]
M5 M4 + aspect × motion, cuts	0.735 [0.678, 0.787]	0.703 [0.643, 0.758]
MT M1 + video type (reference)	0.764 [0.712, 0.813]	0.733 [0.679, 0.783]

Table 13: A7: out-of-fold AUC on the 898 disagreements (95% CIs over clip clusters), with 10 folds grouped by clip cluster, and with the test row's (reader, index) pair also held out of training. MT uses TempCompass's construction label and could not be selected.

A7: development. The rows are the questions where a reader's index and frames answers disagree, for 15 (reader, index) pairs: Haiku on its self-captions, Sonnet captions (224 and 448 px) and Gemini video, frame and frame-clip captions; Sonnet on Sonnet captions (224 and 448 px) and Gemini video captions; Qwen on its self-captions, Haiku's, Sonnet's and Gemini's three. Of 4,455 rows, 898 disagree, and the index is right on 513. The features are all known before the question is read: video (motion; cuts, $\log(1+n)$ for the n consecutive-frame distances above 0.20; the log number of windows), captions of this index (coverage; *mention*, the share of windows whose caption uses a fixed word list for the question's aspect, e.g. for speed "fast", "slowly", "gradually"; 0 for action questions), the task (aspect), and the pair (reader, captioner family). The models are L2 logistic regressions (C = 1). Table 13 gives their out-of-fold AUC; the selection rule (the highest AUC among M1–M5, or the smallest model within 0.01 of it) chose M1. Fit on all 898 rows, its standardized coefficients are: speed −0.57 [−0.93, −0.19] and order +0.30 [−0.04, +0.75] (against action); Sonnet reader −0.60 [−0.88, −0.32] and Qwen reader −0.41 [−0.67, −0.10] (against Haiku); Haiku, Qwen and Sonnet captioners −0.35, −0.53 and −0.22 against Gemini, every CI excluding 0. In decisions, it beats always reading the frames by +0.056 [+0.039, +0.074] and always reading the index by +0.028 [+0.015, +0.041], because it learns per pair which one is better; against that best fixed choice per pair it gains +0.005 [−0.003, +0.014], 0.08 of the headroom to an oracle. Its gains are on the pairs that are close to even: Qwen on Sonnet's captions +0.030 and Haiku's self-captions +0.024 over their best fixed choice.

Aspect	Gemini video	Haiku (self)	Sonnet
action	0.83	0.63	0.76
speed	0.58	**0.33**	**0.47**
direction	0.82	0.61	0.74
order	0.92	0.79	0.87
attribute change	0.84	0.64	0.76

Table 14: The frozen rule for Haiku: P(the index is right | the index and the frames disagree), by aspect and captioner. The rule reads the frames where it is below 0.5 (bold).

A8: the test. The questions are every TempCompass question about the 297 confirmatory videos other than the one the confirmatory set asks: 738 (action 165, direction 163, speed 139, order 139, attribute change 132), 2.5 per video, so all CIs are over clip clusters. Haiku read them on its frames, its self-captions, Sonnet's captions and Gemini's video captions, with the confirmatory prompts. All 2,952 calls answered except one frames read, counted as wrong. Table 14 gives the frozen rule. Of 2,214 (index, question) rows, 492 disagree; the index is right on 102 of 161 for Gemini's captions, 76 of 146 for Sonnet's, and 69 of 185 for the self-captions (in A7 it was right on most disagreements for all three). Within one index the rule varies only by aspect, and its AUC is 0.56 (self-captions), 0.70 (Sonnet) and 0.68 (Gemini); the speed shift is −0.06, −0.24 and −0.17. The accuracy on frames rose from the confirmatory questions mostly on direction (0.55 against 0.28); on the same videos, reworded direction questions give 0.50 against 0.30, while every other aspect matches within 0.02. More of the confirmatory direction questions have four options (49 of 60, against 104 of 163), but this explains almost none of the gap. Weighting the accuracies within each number of options k by both sets' mix of k moves the direction difference from +0.269 [+0.143, +0.391] to +0.261 [+0.129, +0.385], and on the same videos with the same k it is +0.202 [+0.083, +0.324] (99 questions); the new questions score higher both with three options (0.68 against 0.27) and with four (0.48 against 0.29). Over all aspects the shift stays +0.083 [+0.039, +0.128] after weighting by aspect and k (clip-cluster CIs, both sets resampled jointly; predicted in writing before it was computed, research log A10). In hindsight, choosing frames for the self-captions and the index for the other two would score 0.721, above the rule's 0.713, but A7 gave no reason to choose it. Computed after the run, the captioner effect replicates: Sonnet captions − self-captions +0.072, cl. [+0.027, +0.119] (on the confirmatory questions +0.064), and Gemini video captions − self-captions +0.122 [+0.068, +0.177].

H Minimal-pair construction

For each confirmatory question, the sibling is the first video (sorted by ID) built from the same filmed clip that has a TempCompass question with the same aspect, stem and option set and a different correct option. Siblings are the plain and reversed versions of a clip, the other stacked versions (other speed pairs), and the other two-clip videos of the same two clips (other order or layout). Only siblings that Gemini had already captioned were used, which leaves out 3 reversal pairs. This gives 193 pairs: 107 reversal (66 with the reversed video in the confirmatory set, 41 with the plain one), 41 stacked and 45 two-clip, on 174 sibling videos (449 windows). Some siblings are development videos; each sibling item is still a new question–video combination. The sibling is read with the confirmatory question's wording and option order. Chance pair accuracy, both answers right by guessing, is 0.093.

Pair accuracies by group are in Table 2. By group (95% CIs over clip clusters): on stacked pairs, Haiku on Sonnet's captions trails Sonnet on frames by −0.34 [−0.60, −0.08], and Sonnet on its self-captions trails its frames by −0.41 [−0.62, −0.23]; the same-answer rate is above Sonnet on frames' for Haiku on its self-captions (+0.32 [+0.13, +0.49]), Haiku on Sonnet's captions (+0.32 [+0.07, +0.55]) and Sonnet on its self-captions (+0.24 [+0.05, +0.43]), but not clearly for Haiku on Gemini's video captions (+0.22 [−0.02, +0.44]). On two-clip pairs, Haiku's self-captions beat its frames by +0.36 [+0.16, +0.56]. On reversal pairs, Sonnet's captions beat Haiku's self-captions by +0.06 [−0.01, +0.13], where a positive effect was predicted. Haiku on frames gives the same letter on both videos of a pair 54% of the time, and on two-clip pairs its pair accuracy (0.18) is near chance.

I Release

We release the code, every caption, and every reader and judge output (prompt, raw answer, parsed answer and token counts per call), with the log of predictions written before each run. The repository's history, which holds each prediction in a commit dated before its run, is archived by Software Heritage (identifier with-

held for review). Commit dates are set by the author. The open-reader entry was also pushed publicly before its run. An anonymized copy for review is at https://anonymous.4open.science/r/dense_captions-26B/. The videos and questions are TempCompass's (Liu et al., 2024), under CC BY-NC 4.0; we do not redistribute the videos or frames, and the captions, which describe them, are released under the same licence.

J Further related work

Readers on a fixed store. In retrieval-augmented QA, 4.5–83.6% of the gap between the weakest and the strongest reader survives a fixed compressed text across 13 panels (Panthi and Abdelfattah, 2026). In agent memory, three readers tie on a graph store written by a weak model and spread out on one written by a strong model (Peng, 2026), so a readers' tie is not automatic. Ours is on a strong captioner's index.

Text built for the task. Xu and Sankar (2026) turn detector output into text in the task's own vocabulary with fixed code. A text-only model reading it beats a vision-language model of the same size on the image, and the advantage grows as the reader shrinks and reverses at the smallest size tested. The text is engineered for the decision, unlike a question-blind caption, but there too the sign depends on the reader.

What an index keeps. Anonymous (2026a) separate *target inclusion*, whether the queried fact reaches the reader, from *reader recovery*, whether the reader recovers a fact that is present. In these terms, the audit's omissions (§4.4) are failures of inclusion.

Routing. Anonymous (2026b) answer long-video questions from a memory of speech transcripts and captions, and look at the frames only when two derivations of the answer from the text disagree. With the gate held open, looking helps on the questions the gate would escalate and not on the rest. Their signal costs one more text call per question. Ours uses only what is known before reading, and we did not test answer-time signals. EARS (Lin, 2026) routes each video between frame selectors from motion and audio statistics, and, like our scores of motion and cuts, the statistics do not predict which choice is better.

Cost. REEL (Liang and He, 2026) amortizes GPU time over a horizon of questions for one model, and MemoryTax (Peng, 2026) gives a break-even in questions for an agent memory at list prices, with one model writing and reading. Neither has a cheaper reader. For other work on cost, see Zhao et al. (2026); Yammine et al. (2026); Dong et al. (2026); Guan et al. (2026); Kapoor et al. (2024).